

CO 434

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1 Design

1.1 Subspace counting

$V = V(d, q)$, $|V| = q^d$. Then what's the number of 1-dim subspaces? Each 1-dim subspaces contains $q - 1$ non-zero vectors, so there are

$$\frac{q^d - 1}{q - 1}$$

1-dim subspaces.

Definition 1.1. $[k]_q = [k] = \frac{q^k - 1}{q - 1}$. So $V(d, q)$ has $[d]_q$ 1-dim subspaces.

How many 2-dim subspaces? Each 2-dim subspaces contains $[2] = q + 1$ 1-dim subspaces. Count “flags”, pair (1-dim, 2-dim or 1-dim).

$$(\#2 - \dim) \times (q + 1) = [d](\#2 - \dim \text{ on } 1 - \dim)$$

The number is

$$\frac{(q^d - 1)(q^{d-1} - 1)}{(q^2 - 1)(q + 1)} = \frac{[d][d-1]}{[2]}$$

q -binomial Gaussian.

Definition 1.2. We define $[n]!$ recursively by

$$[n + 1]! := [n + 1][n]!$$

which we call q -factorial, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \frac{[n]!}{[k]![n-k]!} = \begin{bmatrix} n \\ n-k \end{bmatrix}$$

Theorem 1.3. The number of k -dim subspaces of $V(d, q)$ is $\frac{[d]!}{[k]![n-k]!}$

Proposition 1.4.

$$\begin{aligned} \begin{bmatrix} d \\ k \end{bmatrix} &= \begin{bmatrix} d \\ d-k \end{bmatrix}, \begin{bmatrix} d \\ k \end{bmatrix} = \frac{[d]}{[k]} \begin{bmatrix} d-1 \\ k-1 \end{bmatrix}, \\ \begin{bmatrix} d \\ k \end{bmatrix}_{q^{-1}} &= q^{-k(n-k)} \begin{bmatrix} d \\ k \end{bmatrix}_q \\ \begin{bmatrix} d \\ k \end{bmatrix} &= q^k \begin{bmatrix} d-1 \\ k \end{bmatrix} + \begin{bmatrix} d-1 \\ k-1 \end{bmatrix} = \begin{bmatrix} d-1 \\ k \end{bmatrix} + q^{d-k} \begin{bmatrix} d-1 \\ k-1 \end{bmatrix} \end{aligned}$$

Remark 1.5. $(x + y)^d = \sum_k \binom{d}{k} x^k y^{n-k}$. If x, y are operators and $yx = qy$, then

$$(x + y)^d = \sum_k \begin{bmatrix} d \\ k \end{bmatrix} x^k y^{d-k}.$$

1.2 Designs

Definition 1.6. An incidence structure $(\mathcal{P}, \mathcal{B})$ is a 2-design if

1. it is point- and block- regular (k points per block and r blocks on a point),
2. for each distinct pair of points, there are exactly λ blocks containing them.

If $|\mathcal{P}| = v$ and $|\mathcal{B}| = b$, then

$$vr = bk, v(v-1)\lambda = bk(k-1)$$

So we find that $\frac{(v-1)\lambda}{r} = k-1$, so

$$\frac{v-1}{k-1} = \frac{r}{\lambda}$$

So we save the parameters (v, b, r, k, λ) to (v, k, λ) , and we call 2- (v, k, λ) design. We will assume $v > k$. The case $k = 1$ is not interesting either.

Example 1.7. 1. Complete designs: all k -subsets of v -set. Here $b = \binom{v}{k}$, $r = \binom{v-1}{k-1}$, and $\lambda =$

2. 2- $(7, 3, 1)$. $\mathcal{P} = \{0, \dots, 6\}$, blocks $\{(0, 1, 3), (1, 2, 4), \dots\}$

3. 2- $(9, 3, 1)$, 2- $(13, 3, 1)$.

Theorem 1.8 (Wilson's). If k and λ are given, there is a constant v_0 such that if $v \geq v_0$, there is a 2- (v, k, λ) design.

Definition 1.9. A block design is a Steiner system if $\lambda = 1$. Notice that a Steiner system is a partial linear space. (in general, it is not a dual linear space)

Example 1.10. 1. 2- $(v, 2, 1)$,

2. 2- $(v, 3, 1)$, $v = 7$ works. Steiner triple system.

3. 1-dim versus 2-dim subspaces of $V(3, q)$ (projective plane) 2- $(q^2 + q + 1, q + 1, 1)$.

We will focus on Steiner triple systems, i.e., 2- $(v, 3, 1)$ designs. From earlier result,

$$\begin{aligned} \frac{v-1}{k-1} = \frac{r}{\lambda} = r &\implies r = \frac{v-1}{2} \\ b = \frac{v(v-1)}{k(k-1)} &= \frac{v(v-1)}{6} \end{aligned}$$

The former tells us v must be odd, and the latter tells us $v \equiv 0, 1, 3, 4 \pmod{6}$. Therefore if a triple system exists $v \equiv 1, 3 \pmod{6}$.

v	b
1	0
3	1
7	7 $\leftarrow$ (0,1,3) and its cyclic shifts
9	12 $\leftarrow$ product
13	26
15	35

A product construction:

Let S be a triple system on v points. Take three copies.

Definition 1.11. A subspace of a partial linear space is a subset S of points such that l is a line and $|l \cap S| \geq 2$ then $\overline{l \cap S} \subseteq S$.

Example 1.12. Any points or any block is a subspace.

Example 1.13. Assume S is a $2-(v, 3, 1)$ design. How big can S be? In fact, it is at most $\frac{v-1}{2}$. (For a fixed point outside S , and any point in S , a line crossing them, then there must be another point outside S that is in the line).

Let r_S be the value of r for S .

$$v - |S| = (r - r_S)(k - 1) \implies v = 2S + 1$$

So we get a 1-factorization of K_{2S+1} . Now, we are using $k = 3$, each point in S determines a partition of the $S + 1$ points not in S into pairs.

How do we construct 1-factorization of K_{2n} ? For a $\{0, \dots, 6\} \times \{0, \dots, 6\}$ square, the rule is $\frac{1}{2}(x + y)$, this gives a symmetric idempotent (the i^{th} entry in the diagonal is i) Latin square.

$$\begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & & 0 & 4 & 1 & 5 & 2 & 6 & 3 \\ 1 & & 4 & 1 & 5 & 2 & 6 & 3 & 0 \\ \dots & & & & & & & & \end{array}$$

Theorem 1.14. For each n , there is a bijection between the 1-factorizations of K_{2n} and symmetric idempotent Latin squares of order $2n - 1$.

Theorem 1.15 (Bose). If there is a symmetric idempotent Latin square on $2v + 1$ points, there is a Steiner triple system on $6v + 3$ points.

Proof. Outline of proof when $m = 5$:

$$\begin{array}{ccccc} 1 & \text{---} & 1 & \text{---} & 1 \\ & \swarrow & & \searrow & \\ 2 & \text{---} & 2 & \text{---} & 2 \\ & \swarrow & & \searrow & \\ 3 & \text{---} & 3 & \text{---} & 3 \\ & \swarrow & & \searrow & \\ 4 & \text{---} & 4 & \text{---} & 4 \\ & \swarrow & & \searrow & \\ 5 & \text{---} & 5 & \text{---} & 5 \end{array}$$

$L_{i,j} = \frac{1}{2}(i + j)$, i, j in column 1, and $L_{i,j}$ in column two. □

Remark 1.16. 1. If $v \equiv 3 \pmod{6}$, there is an STS on v points (Bose).

2. If $v \equiv 1 \pmod{6}$, there is an STS on v points (Skolem).
3. For each $v \equiv 1, 3 \pmod{6}$, there is a cyclic STS.
4. A parallel class in a design is a set of blocks that partition the points.
5. A design is resolvable if the blocks can be partitioned into parallel classes.
If a triple system is resolvable, then $3|v$. This condition is sufficient.
6. Large sets: can we partition the set of all triples for $\{1, \dots, v\}$ into copies of a given Steiner triple system?

1.3 Point Graphs of Steiner Triple Systems

Let S be a Steiner system with incidence matrix N , then

$$NN^T = (r - \lambda)I + \lambda J$$

Note that if N is the incidence matrix of an incidence structure S and $NN^T = cI + dJ$, then S is a 2 -(v, k, λ) design.

Next,

$$NN^T = c_0A_0 + \dots + c_kA_k, A_0 = I, A_i \text{ symmetric}, \sum A_i = J$$

If S is a Steiner system,

$$NN^T = (r - 1)I + J$$

and

$$NN^T = \begin{bmatrix} k & & 01 \\ & \ddots & \\ 01 & & k \end{bmatrix} = kI + A_1, A_1 \text{ is the blocks with a point in common}$$

Where A_1 is the adjacency matrix of a graph on the blocks of the Steiner system. The eigenvalues of $(r - \lambda)I + \lambda J$ are $r - \lambda^{(v-1)}, r - \lambda + v\lambda^{(1)}$, hence it is invertible.

Theorem 1.17 (Fisher's inequality). If $\mathcal{B}$ is a 2 -(v, k, λ) design, then $b \geq v$.

Proof. Since $NN^T = (r - \lambda)I + \lambda J$ is invertible, if $z^T NN^T = 0$, then $z = 0$. So if there is a z such that $z^T N = 0$, then $z^T NN^T = 0 \implies z = 0$. So the rows of N are linearly independent, hence $v \leq b$. $\square$

Definition 1.18. A graph is strongly regular if there are constants (v, k, a, c) such that

- The graph has v vertices.
- It is k -regular.
- If u, v are adjacent vertices, then u, v have exactly a common neighbours.
- If u, v are distinct and not adjacent, they have c common neighbours.

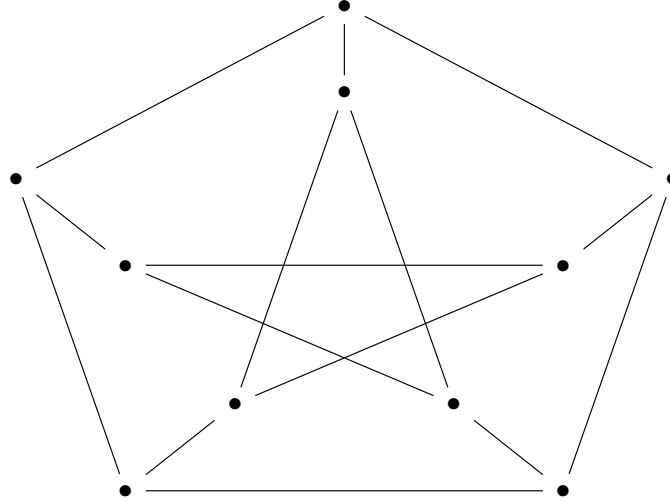
- The graph is not complete or empty.

Fix a vertex u , then it has k neighbours and there are $(v - 1 - k)$ vertices left (at distance 2 from u). The k -neighbours form a a -regular graphs on k vertices.

Proposition 1.19. The complement of a strongly regular graph is strongly regular.

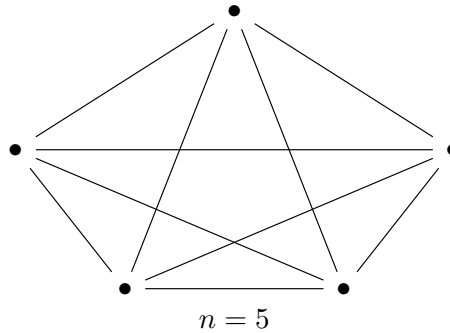
Example 1.20. • mK_n and $\overline{mK_n}$ (the complement of mK_n). These are referred to as imprimitive strongly regular graphs.

- Peterson graph



$$v = 10, k = 3, a = 0, c = 1.$$

- $C_5, (5, 2, 0, 1)$.
- $K_{m,m}$ complete bipartite graph.
- $L(K_n)$, the line graph of K_n



If X is a graph, the vertices of the line graph $L(X)$ are its edges. Two edges are adjacent in $L(X)$ if they have exactly one vertex in common. $\overline{L(K_5)}$ is the Peterson graph.

v		val	$v - 1 - k$		#
9	1	9	2	12	1
13	1	15	10	26	2
15	1	18	16	35	80
19	1	24	32	57	11,084,874,829

Table 1: Caption

1.4 Linear Algebra

Theorem 1.21. If C and D are matrices and both CD and DC are defined. Then

$$\det(I - CD) = \det(I - DC)$$

Proof. We have

$$\begin{pmatrix} I & C \\ D & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -D & I \end{pmatrix} = \begin{pmatrix} I - CD & C \\ 0 & I \end{pmatrix}$$

$$\begin{pmatrix} I & 0 \\ -D & I \end{pmatrix} \begin{pmatrix} I & C \\ D & I \end{pmatrix} = \begin{pmatrix} I & C \\ 0 & I - DC \end{pmatrix}$$

The determinants of the LHS are equals, consequently $\det(I - CD) = \det(I - DC)$ □

Corollary 1.22. $\det(I - xCD) = \det(I - xDC)$.

Corollary 1.23. If C is $m \times n$ and D is $n \times m$, then

$$x^n \det(I - CD) = x^m \det(I - DC)$$

Corollary 1.24 (over $\mathbb{R}$). CD and DC have the same eigenvalues with the same multiplicities.

Example 1.25. Let S be a Steiner triple system, then

$$NN^T = (r - 1)I + J, N^T N = kI + A$$

$$(r - 1)I + J = (r - 1)\left(I + \underbrace{\frac{1}{r - 1}\mathbf{1}}_C \underbrace{\mathbf{1}^T}_D\right)$$

By the theorem,

$$\det\left(I + \frac{1}{r - 1}J\right) = \det\left(I_1 + \frac{1}{r - 1}\mathbf{1}\mathbf{1}^T\right)$$

The block graph of a Steiner system is strongly regular.

For a triple system $2-(v, 3, 1)$, $r = \frac{v-1}{2}$, $b = \frac{v(v-1)}{6}$, $n \equiv 1, 3 \pmod{6}$. The block graph has $\frac{v(v-1)}{6}$ vertices, split up as follows:

a fixed vertex a , it has $\frac{3(v-3)}{2}$ neighbours, and $\frac{(v-3)(v-7)}{6}$ vertices of distance 2. $c = 9, a = ?$

1.5 Symmetric Designs and Projective Planes

Theorem 1.26. $\mathcal{D}$ is a symmetric (v, k, λ) design, then any two distinct blocks have exactly λ points in common.

Proof. We have $NN^T = (r - \lambda)I + \lambda J$, and N is invertible. So

$$\begin{aligned} N^T N &= N^{-1} N N^T N \\ &= N^{-1}((r - \lambda)I + \lambda J)N \\ &= N^{-1}((r - \lambda)N + \lambda k N) \\ &= (r - \lambda)I + \lambda k \underbrace{N^{-1}J}_{\text{constant}} \end{aligned}$$

This shows that the dual of a symmetric design is symmetric. Since N^T is the incidence matrix of a symmetric design, any two distinct blocks of the original design meet in λ points. $\square$

Definition 1.27. If $\mathcal{D}$ is a symmetric $2-(v, k, \lambda)$ design and β is a block in $\mathcal{D}$, then the blocks sets $\{\alpha \setminus \beta : \alpha \text{ a block}, \alpha \neq \beta\}$ is a 2-design. This is called a residual design.

If we apply this to a projection plane, we get an affine plane.

Questions

1. A projective plane is a symmetric design with parameters $2-(q^2 + q + 1, q + 1, 1)$. If such a design exists, must q be a prime power?
2. Consider the symmetric designs with parameters $2-(v, k, \lambda)$ with λ fixed. Are there infinitely many such designs? What about $\lambda = 2$?

Example 1.28. 1. $2-(11, 5, 2)$ with 11 blocks $(1 \ 3 \ 4 \ 5 \ 9), (2 \ 4 \ 5 \ 6 \ 10), \dots$

2. 2-dim subspaces and $(d - 1)$ -dim subspaces of $V(d, q)$.

Theorem 1.29. If $U_1, U_2 \leq V$, then $\dim(U_1 + U_2) = \dim(U_1) + \dim(U_2) - \dim(U_1 \cap U_2)$

It follows that we get a symmetric design as $\dim(U_1 \cap U_2) \neq 0$ by the above theorem. If $d = 4$, then we have a projective plane.

3. If $\mathcal{D}$ is a $2-(v, k, \lambda)$ design with point set V_1 , then the sets

$$\{V \setminus \alpha : \alpha \text{ is a block}\}$$

form a 2-design. This is the complement of $\mathcal{D}$. So the complement of a symmetric design is symmetric.

If a symmetric $2-(v, k, \lambda)$ design exists, then

$$\lambda v(v - 1) = bk(k - 1)$$

$$v - 1 = \frac{k^2 - k}{\lambda}$$

$$v = \frac{k^2 - k}{\lambda} + 1 \implies \lambda | (k^2 - k)$$

Theorem 1.30 (Brnek-Ryser-Chowla). Assume that a symmetric 2 -(v, k, λ) design exists, then

1. If v is even, then $k - \lambda$ is a square.
2. If v is odd, the equation

$$(k - \lambda)x^2 + (-1)^{\frac{v-1}{2}}\lambda y^2 = z^2$$

has a non-zero solution in integers.

The difference $r - \lambda$ is called the order of the design. So the projective plane with $q^2 + q + 1$ points has order q .

(a). If

$$\hat{N} = \begin{pmatrix} N & \mathbf{1} \\ \lambda \mathbf{1}^T & k \end{pmatrix}$$

then

$$\hat{N} \begin{pmatrix} I_v & 0 \\ 0 & -\lambda \end{pmatrix} \hat{N}^T = n \begin{pmatrix} I_v & 0 \\ 0 & -\lambda \end{pmatrix}$$

Hence

$$\det(\hat{N})^2(-\lambda) = n^{v+1}(-\lambda)$$

So if v is even, then n is a square. □

Sums of two squares

If $x \in \mathbb{Z}$, then $x^2 \equiv 0, 1 \pmod{4}$ and if $x, y \in \mathbb{Z}$, then $x^2 + y^2 \equiv 0, 1, 2 \pmod{4}$. Hence if $m \equiv 3 \pmod{4}$, it cannot be written a sum of two squares.

If $m = a^2 + b^2, a, b \in \mathbb{Z}$, then

$$m = (a + ib)(a - ib)$$

Here, if $a, b \in \mathbb{Z}$, then $a + ib$ is a Gaussian integer. The Gaussian integers form a commutative ring. If $n = (c + id)(c - id)$, then we have

$$mn = (ac - bd)^2 + (bc + ad)^2$$

Theorem 1.31. If p is a prime and $p \equiv 1 \pmod{4}$, then p is the sum of two squares.

Theorem 1.32. Assume $n = p_1^{\mu_1} \cdots p_n^{\mu_n}$, then n is the sum of two squares if and only if whenever $p_i \equiv 3 \pmod{4}$, μ_i is even.

Corollary 1.33. If n is the sum of two rational squares, then its the sum of two integers.

Missing material

1. equivalence of quadratic forms

2. Witt's cancellation theorem
3. Gaussian integers and sums of two squares
4. Quaternions and sums of four squares
5. BRC and Legendre

Applications Projective planes $2-(n^2 + n + 1, n + 1, 1)$ designs. By BRC, if n is even, then

$$nx^2 + (-1)^{\binom{n}{2}}y^2 = z^2$$

has a non-zero integer solution. If $n \equiv 0, 3 \pmod{4}$, then $v \equiv 1 \pmod{4}$, and $nx^2 + y^2 = z^2$, where $(0, 1, 1)$ is a solution. If $n \equiv 1, 2 \pmod{4}$, then $v \equiv 1 \pmod{4}$ and $nx^2 = y^2 + z^2$, then

$$n = \left(\frac{y}{x}\right)^2 + \left(\frac{z}{x}\right)^2 \implies n \text{ is a sum of two squares}$$

Suppose $n = 10$, parameter $(111, 11, 1)$ and BRC gives

$$10x^2 + y^2 = z^2$$

Then $(0, 1, 1)$ is a solution. Despite this, there is no projective plane of order 10. (C.Lam)
 $\lambda = 2$ (biplanes):

1. $(22, 7, 2)$ v is even, n is not square
2. $(29, 8, 6)$ fails $6x^2 + 2y^2 - z^2 = 0$

$$\implies 3x^2 + y^2 - 2z^2 = 0 \text{ NO! Legendre}$$

1.6 Projective and Affine Planes

Definition 1.34. In terms of designs, a finite projective plane is a symmetric $2-(n^2 + n + 1, n + 1, 1)$ design. More generally, an incidence structure $\mathcal{D}$ is a projective plane if

1. any two distinct points lie on a unique line,
2. any two distinct lines have exactly one common point,
3. there are four points, no three on a line. Equivalently, the minimum valency of the incidence graph is ≥ 3 .

Definition 1.35. Another approach: an incidence structure is a projective plane if its incidence graph has diameter 3, girth six, and minimal valence ≥ 3 .

Example 1.36. K_6 has 15 perfect matchings. The 15 edges and 15 perfect matchings form an incidence structure, its incidence graph is bipartite, 3-regular, on 30 vertices. This has diameter four and girth eight.

Definition 1.37. A finite affine plane is a $2-(q^2, q, 1)$ -design (a Steiner system).

Definition 1.38. Geometrically, an affine plane is an incidence structure such that:

1. it is a partial linear space;
2. if l is a line and x is a point not in l , there is a unique line on x disjoint from l ;
3. there exists three points **not all** on the same line.

Example 1.39. 2-(4,2,1), 2-(9,3,1), 2-(16,4,1). Notice that 2-(9,3,1) is a Steiner triple system.

Remark 1.40. 1. The points and cosets of the 1-dim subspaces of $V(2, \mathbb{F})$ form an affine plane.

2. In the above definition, the unique line through x that is disjoint from l is said to be parallel to l . This is an equivalence relation on the lines.
3. The full parameter set for an affine plane is

$$(q^2, q^2 + q, q + 1, q, 1)$$

Recall that $k = r - \lambda$ is the order of a design. In this case, it is of order q .

One may show that two definitions are equivalent.

Proof. Exercise. □

1.7 Residual Designs

Definition 1.41. Assume $\mathcal{D}$ is a symmetric 2- (v, k, λ) -design. If β is a block in $\mathcal{D}$, then the residual design relative to β has

1. its points are the points of $\mathcal{D}$ not on β .
2. if α is a block of $\mathcal{D}$, then $\alpha \setminus \beta$ is a block of the residual design.

Since $\mathcal{D}$ is symmetric, any two distinct blocks have λ points in common, so the residual design has parameters:

$$(v - k, v - 1, r, k - \lambda, \lambda)$$

Example 1.42. 2- $(q^2 + q + 1, q + 1, 1)$ gives 2- $(q^2, q, 1)$. So affine planes are residuals of projective planes.

2-(11,5,2) gives 1-(10,3,2) design. The parameters of a biplane with block size k are:

$$(\frac{k^2 - k}{2} + 1, k, 2)$$

and a residual design has parameters

$$(\frac{k^2 - 3k + 2}{2}, k - 2, 2)$$

order	v	k
2	7	4
3	11	5
4	16	6
	$\vdots$	

Theorem 1.43. Every affine plane is the residual of a projective plane.

Theorem 1.44. $\mathcal{P}$ is a 2-design with the parameters of a residual biplane, it is a residual.

1.8 Affine Planes as Residuals

Given a $2-(n^2, n, 1)$ -design. Each parallel class consists of $n + 1$ sets of n pairwise disjoint lines.

1.9 Hadamard matrices

Definition 1.45. A Hadamard matrix H is an $n \times n$ matrix with entries ± 1 such that $H^T H = nI$. Then $\frac{1}{\sqrt{n}}H$ is orthogonal.

Example 1.46.

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$$

If P_1, P_2 are permutation matrices and D_1, D_2 are diagonal matrices, with diagonal entries ± 1 , then

$$P_1 D_1 H D_2 P_2 \text{ and } H$$

are equivalent. A matrix of the form PD is monomial.

A Hadamard matrix is normalized if all entries in its first row and first column are 1. If $e_1^T H$, the first row of H , is $\mathbf{1}^T$, what is the second row? It must have the same number of 1's and -1's. This implies n must be even. Now, consider the third row $e_3^T H$. This is orthogonal to $(e_1 \pm e_2)^T H$, and therefore $4|n$.

Question: Is it true that if $4|n$, there is an $n \times n$ Hadamard matrix? The first open case is 668.

1.10 Kronecker Products

Definition 1.47. If A & B are matrices, then the Kronecker product $A \otimes B$ is the block matrix with the ij -block $A_{ij}B$.

Example 1.48.

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Proposition 1.49. 1. If the required products exist, we have

$$(A \otimes B)(C \otimes D) = AC \otimes BD$$

Hence

$$(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$$

$$2. (A \otimes B) \otimes C = A \otimes (B \otimes C)$$

$$3. A \otimes (B + C) = A \otimes B + A \otimes C$$

Definition 1.50. We define the Schur product of two $m \times n$ matrices A and B to be

$$(A \circ B)_{ij} = A_{ij} \cdot B_{ij}$$

Then, for example,

$$A \otimes (B \circ C) = (A \otimes B) \circ (A \otimes C)$$

If P is the operator on the vector space $U \otimes V$ that sends $(u \otimes v)$ to $(v \otimes u)$, then $P^2 = I$ and P is a permutation matrix. We call this the flip operator.

Remark 1.51. $P(A \otimes A^T)$ is symmetric.

Question: $(A \otimes B)^T = A^T \otimes B^T$. Yes!

Theorem 1.52. If $\mathcal{D}$ is a symmetric 2 -(v, k, λ) design, then

$$4n - 1 \leq v \leq n^2 + n + 1$$

where $n = r - \lambda$ is the order of the design.

of lower bound. Let N be the incidence matrix of $\mathcal{D}$. Each entry of $2N - J$ is ± 1 and $(2N - J)\mathbf{1} = (v - 2k)\mathbf{1}$. So

$$\mathbf{1}^T(2N - J)^T(2N - J)\mathbf{1} = v(v - 2k)^2$$

On the other hand,

$$\mathbf{1}^T(2N - J)^T(2N - J)\mathbf{1} = \mathbf{1}^T(4N^T N - 4NJ + vJ)\mathbf{1} = 4kv^2 + v^2(v - 4k)$$

Hence $v^2 - 4nv + 4n^2 = 4n^2 - 4n$. If $n \geq 2$, then $4n^2 - 4n$ is not a square, therefore

$$(v - 2n)^2 \geq (2n - 1)^2$$

□

Theorem 1.53. Suppose $\mathcal{D}$ is symmetric and $v = 4n - 1$. Define $\overline{N} = 2N - J$. Then

$$\begin{bmatrix} 1 & 1^T \\ 1 & \overline{N} \end{bmatrix}$$

is a normalized Hadamard matrix.

If $H = \begin{pmatrix} 1 & 1^T \\ 1 & H_1 \end{pmatrix}$ is Hadamard, then $\frac{1}{2}(H_1 + J)$ is a symmetric design with parameters one of the following:

1. $(4n - 1, 2n, n)$
2. $(4n - 1, 2n - 1, 2n - 1)$

There is a second construction of designs from Hadamard matrices. Assume H is Hadamard and its first row is 1. Then each row than the first splits into two “blocks” of size n . We get a design with line $2m$ columns of H as points, and the $4m - 2$ blocks as blocks. If H is $n \times n$, the parameters of this design are

$$(n, 2n - 2, n - 1, \frac{n}{2}, \frac{n}{2} - 1)$$

They are called Menon designs.

2 Special classes of Hadamard designs

2.1 Regular Hadamard matrices

Definition 2.1. A Hadamard matrix H is regular if all row sums are equal, equivalently $H\mathbf{1} = s\mathbf{1}$ for some s . This implies the column sums are equal: we have

$$\begin{aligned} n\mathbf{1} &= H^T H\mathbf{1} = sH^T\mathbf{1} \\ \implies H^T\mathbf{1} &= \frac{n}{s}\mathbf{1} \end{aligned}$$

Consequently, $\frac{n}{s} = s$, so $n = s^2$.

Lemma 2.2. If H is regular $n \times n$ then n is a square and $s = \pm\sqrt{n}$.

If H_1, H_2 are regular, so is $H_1 \otimes H_2$.

Lemma 2.3. Let H be an $n \times n$ matrix with entries ± 1 . Then H is a regular Hadamard matrix if and only if $\frac{1}{2}(J - H)$ is the incidence matrix of a symmetric design with parameters $2-(4h^2, 2h^2 - h, h^2 - h)$.

2.2 Symmetric Hadamard matrices

Assume H is Hadamard $n \times n$. Then $\frac{1}{\sqrt{n}}H$ is orthogonal and so its eigenvalues. Note that $HH^T = H^TH$, H is normal and therefore there is a unitary matrix U such that $H = UDU^*$ and D is diagonal. Further $D = U^*HU$ is unitary,

H is symmetric, $H^2 = nI$ and the minimal polynomial of H is $t^2 - n$. Therefore the eigenvalues of H are $\pm\sqrt{n}$.

Lemma 2.4. If H is a symmetric Hadamard matrix with constant diagonal, then n is a square.

Assume H is symmetric with a constant diagonal. Replace H by $-H$ if needed. We may assume $H \circ I = I$ (that is, the diagonal is 1). Suppose H has exactly a positive eigenvalues, then

$$n = \text{tr}(H) = a\sqrt{n} + (n - a)(-\sqrt{n}) = (2a - n)\sqrt{n}$$

This implies $\sqrt{n}$ is an integer.

Remark 2.5. If H is Hadamard, then $P(H \otimes H^T)$ is symmetric, with constant diagonal.

3 Conference matrices

A conference matrix is a $(0, \pm 1)$ matrix, order $n \times n$, C such that $C^T C = (n-1)I$. We assume that $C \circ I = 0$. Two conference matrices C_1 and C_2 are equivalent if

$$C_2 = P D_1 C_1 D_2 P^T$$

A conference matrix is normalized if all entries in its first row and column are non-negative.

Theorem 3.1. Assume $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & S \end{pmatrix}$ is a conference matrix. If $n \equiv 2 \pmod{4}$ then S is symmetric; if $n \equiv 0 \pmod{4}$, S is skew-symmetric.

Proof. If $C^T = -C$, then

$$\begin{aligned} (C^T + I)(C + I) &= C^T C + C + C^T + I \\ &= (n-1)I + C - C + I \\ &= nI \end{aligned}$$

So $C + I$ is Hadamard. □

Theorem 3.2. Let C be conference. Then C is equivalent to $\begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & S \end{pmatrix}$. If $n \equiv 2 \pmod{4}$, then S is symmetric. If $n \equiv 0 \pmod{4}$, then S is skew-symmetric.

Proof. Let c_1, c_2, c_3 denote the first three rows of C . Then

$$(c_1 + c_2) \circ (c_1 + c_3) = c_1 \circ c_1 + c_2 \circ c_1 + c_1 \circ c_3 + c_2 \circ c_3$$

If we write C as

$$\begin{pmatrix} 0 & 1 & 1 & \cdots \\ 1 & 0 & x & \cdots \\ 1 & y & & \end{pmatrix}$$

Recall that $\langle x, y \rangle = \mathbf{1}^T x \circ y$. The inner product of the left side with $\mathbf{1}$ is $3 + x + y + 4m$ for some m , and the inner product of the right side with $\mathbf{1}$ is $n-1$. So

$$n = 4 + x + y + 4m \implies n \equiv x + y \pmod{4}$$

and hence

$$x + y \equiv \begin{cases} 2 \pmod{4} & \text{if } n \equiv 2 \pmod{4} \\ 0 \pmod{4} & \text{if } n \equiv 0 \pmod{4} \end{cases}$$

This result is proved by Goethals and Seidel. □

Lemma 3.3. If C is skew, $I + C$ is Hadamard.

Proof. $(I + C)(I + C)^T = I + C + C^T + CC^T = I + C - C + (n-1)I = nI$ □

Lemma 3.4. If C is symmetric, then $\begin{pmatrix} C+I & C-I \\ I-C & C+I \end{pmatrix}$ is Hadamard.

Proof. bookkeeping. □

Lemma 3.5. If C is $n \times n$ conference and $n \equiv 2 \pmod{4}$, then n is the sum of two squares

4 Quadratic forms in odd characteristic

If V is a vector space and $Q(x)$ is an $\mathbb{F}$ -value function on V , it is a quadratic form if

1. $Q(ax) = a^2Q(x), a \in \mathbb{F}, x \in V$
2. $Q(x+y) - Q(x) - Q(y)$ is bilinear.

In practice,

$$Q(x) = \sum_{i,j} a_{i,j} x_i x_j \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

If A is symmetric, $x^T A x$ is a quadratic form.

Two quadratic forms $x^T A x$ and $x^T B x$ are equivalent if there is a matrix G over $\mathbb{F}$ with determinant 1 such that $B = G^T A G$.

Sum: if Q_1 is a quadratic form on V_1 and Q_2 is a quadratic form on V_2 . Then we define $Q_1 \oplus Q_2$ on $V_1 \oplus V_2$ by

$$(Q_1 \oplus Q_2)(v_1 \oplus v_2) := Q_1(v_1) + Q_2(v_2)$$

Theorem 4.1. If $Q_1 \oplus Q_2$ is equivalent to $Q_1 \oplus Q_3$, then Q_2 and Q_3 are equivalent. This is called Witt cancellation.

We write $Q_1 \simeq Q_2$ to denote that Q_1, Q_2 are equivalent.

Now back to conference matrices. We have

$$CC^T = (n-1)I$$

and $n \equiv 2 \pmod{4}$. The above equation implies that $I_n \simeq (n-1)I_n$. We also have that $nI_4 \simeq I_4$ (Lagrange sum of four squares). So

$$CC^T = \begin{pmatrix} I_4 & & \\ & \ddots & \\ & & I_4 \\ & & & (n-1)I_4 \end{pmatrix}$$

and hence $(n-1)I_2 \simeq I - 2$. So

$$\begin{pmatrix} n-1 & 0 \\ 0 & n-1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} I \begin{pmatrix} a & c \\ b & d \end{pmatrix} \implies n-1 = a^2 + b^2$$

Now, we construct some conference matrices.

Let $\mathbb{F}$ be a finite field of order q . Define a function χ on $\mathbb{F}$ by

$$\chi(x) = \begin{cases} 1 & x \text{ is a square on } \mathbb{F} \\ -1 & \end{cases}$$

We say χ is a quadratic character on $\mathbb{F}$. Note that $\chi(x) \equiv x^{\frac{q-1}{2}}$. Further, $\chi^2 = 1$ on $\mathbb{F}^\times$.

Example 4.2. $q = 5$:

$$\begin{array}{c|ccccc} 0 & 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{array} \quad \chi(b-a)$$

Denote this matrix by P . It is a Jacobsthal matrix (P is $(q-1) \times (q-1)$)

Theorem 4.3. $PP^T = qI - J$.

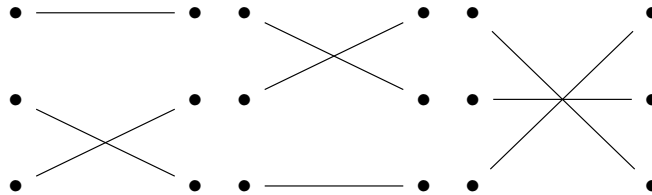
5 Orthogonal Arrays

Latin squares are $n \times n$ matrices with each row, column entries to be $[n]$.

Example 5.1. For $n = 3$, we have

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 3 \\ 0 & 3 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

where the three matrices on the RHS corresponding to respectively



which are edge 3-colouring of $K_{3,3}$.

Example 5.2. A multiplication table for a finite group of order n is a Latin square.

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad \mathbb{Z}_3$$

Definition 5.3. An orthogonal array $OA(k, n)$ is an $n^2 \times k$ array with entries $1, \dots, n$ such that each pair (i, j) occurs exactly once in each pair of columns.

Example 5.4. For the multiplication table for $\mathbb{Z}_3$,

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \\ 1 & 0 & 1 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \\ 2 & 1 & 0 \\ 2 & 2 & 1 \end{pmatrix}$$

Example 5.5. An $OA(3, n)$ is an $n \times n$ Latin square. An $OA(4, n)$ is a pair of orthogonal Latin squares.

The graph of an orthogonal array $OA(k, n)$ has the n^2 rows as vertices, with two vertices adjacent if they agree on one coordinate. Notice that two distinct k -tuples cannot have more than two coordinates that agree. Two orthogonal arrays are equivalent if their graphs are isomorphic. Construction: $\mathbb{F} = GF(q)$.

$$(x \ y \ x + \lambda_1 y \ \cdots \ x \lambda_m) \quad x, y \in \mathbb{F}, \lambda_i \in \mathbb{F} \setminus \{0\} \text{ distinct}$$

If $\mathcal{D}$ is a $2-(v, 3, 1)$ design, its graph has the blocks as vertices, blocks adjacent if they have exactly one point in common. This is strongly regular, there are parameters (v, k, a, c) such that the graph is k -regular on v vertices, two adjacent vertices have a common neighbours and two distinct non-adjacent vertices have c common neighbours. What are the parameters?

# of vertices	$\frac{v(v-1)}{6}$
valency	$\frac{3v-9}{2}$
a	$\frac{v+3}{2}$
c	9

If X is (v, k, a, c) strongly regular, then

$$\begin{aligned} A^2 &= kI + aA + c(J - I - A) \\ \implies A^2 - (a-c)A - (k-c)I &= cJ \end{aligned}$$

Then

$$\begin{aligned} cJ\mathbf{1} &= (A^2 - (a-c)A - (k-c)I)\mathbf{1} = (k^2 - k(a-c) - (k-c))\mathbf{1} \\ \implies cv^2\mathbf{1} &= (k^2 - ak + ck - k + c)\mathbf{1} \end{aligned}$$

Therefore we know k, a, c determines v and $\mathbf{1}$ is an eigenvector. For other eigenvectors that orthogonal to $\mathbf{1}$, we have $Jz = 0$ and $Az = \lambda z$ for some λ . Then we obtain

$$0 = (A^2 - (a-c)A - (k-c)I)z = \lambda^2 z - \lambda(a-c)z - (k-c)z$$

$$\implies \lambda^2 - (a - c)\lambda - (k - c) = 0$$

So we get

$$\lambda = \frac{1}{2}(a - c \pm \sqrt{(a - c)^2 + 4(k - c)})$$

For a Steiner triple system,

$$\begin{aligned} a - c &= \frac{v - 1}{2} \\ k - c &= \frac{3(v - 9)}{2} \\ \lambda^2 - \frac{v - 15}{2}\lambda - \frac{3(v - 9)}{2} &= (\lambda + 3)(\lambda - \frac{v - 9}{2}) \end{aligned}$$

Definition 5.6. A parallel class in a semiregular incidence structure is a set of blocks that partition the vertices. If a Steiner triple system has a parallel class, then $3|v$.

6 A 1-factorization of K_n

Assume $n = 2m$ and $V(k) = \{\infty, 0, 1, \dots, 2m - 2\}$. For $i = 0, \dots, 2m - 2$, let the ∞i entry to be i , i, j entry to be $\frac{1}{2}(i + j)$. Gives a Latin square of order $2m - 1$

$$L_{ij} = \frac{1}{2}(i + j)L_{i,i} = i$$

This Latin square is symmetric and idempotent ($L_{i,i} = i$). Let $\mathcal{D}$ be a 2 -($v, 3, 1$) design with blocks $(i, j, i \star j)$. Define $i \star i = i$,

$$L_{ij} = \begin{cases} i \star j & , i \neq j \\ i & , i = j \end{cases}$$

Converse: Let L be a symmetric idempotent square Latin square of order $n \times n$ (so n is odd). Set $N = \{1, \dots, n\}$. We construct a Steiner triple system on $V = N \times \{1, 2, 3\}$ with triples

$$\begin{aligned} ((v, i), (s, i), L_{r,s}, i + 1) & - 3n(n - 1)/2 \\ ((r, 0), (r, 1), (r, 2)) \end{aligned}$$

7 Reconstructing a Steiner triple system from its block graph

Let $\mathcal{D}$ be a 2 -($v, 3, 1$) design with block graph X . The blocks on a point of the design form a clique of size $\frac{v-1}{2}$. What's the maximal size of a clique?

Claim: If $v > 15$, then the maximal clique size is r , and any clique of that size consists of the triples on a point.

8 Block Intersections

Definition 8.1. The degree set of a collection of k -subsets of a set is

$$\{|\alpha \cap \beta| : \alpha \neq \beta\}$$

The degree of a subset is the size of the degree set.

Remark 8.2. The degree of a Steiner triple system is 2, and the degree set is $\{0, 1\}$

Definition 8.3. A 2-design is quasisymmetric if it has degree two.

Example 8.4 (quasisymmetric). 1. Steiner systems

2. residuals of biplanes (e.g., $2-(11, 5, 2) \mapsto 2-(6, 3, 2)$)

3. Affine resolvable designs

4. multiple of a symmetric design

Theorem 8.5. The block graph of a quasisymmetric design is strongly regular.

Remark 8.6. We usually denote the degree set of a quasisymmetric design by $\{x, y\}$ with $x < y$.

Let N be the incidence matrix of a quasi-symmetric design. Then

$$N^T N = kI + xA + y(J - I - A) = (k - y)I + (x - y)A + yJ$$

$$NN^T = (r - \lambda)I + \lambda J$$

Notice that NN^T and $N^T N$ has the same non-zero eigenvalues with the same multiplicity. Notice NN^T has eigenvalues $(r - \lambda)^{(v-1)}, bk^{(1)}$. It follows that A is k -regular, connected (wlog), and has exactly three eigenvalues. Denote them by k, θ, τ with $(\tau < 0 \leq \theta)$ and so the minimal polynomial is $(t - k)(t - \theta)(t - \tau) \implies (A - \theta I)(A - \tau I) = \gamma J$ for some γ . Now this implies

$$A^2 - (\theta + \tau)A + \theta\tau I = \gamma J$$

An incidence structure is a 1-design if it is point- and block-regular. A trivial example is any parallel classes.

Theorem 8.7. Let $\mathcal{D}$ be a $2-(v, b, r, k, \lambda)$ design. If there is a partition π of the blocks into 1-designs, then

$$b \geq v + |\pi| - 1$$

Proof. Let N be the incidence matrix of $\mathcal{D}$, and let

$$\pi = (R_1, R_2, \dots, R_c)$$

be a partition of the blocks into 1-designs. The columns indexed by R_i sum to a multiple of $\mathbf{1}$, for each i . If N' is the matrix we get by deleting one column from each R_i , and adjoining a column $\mathbf{1}$. Then the rank does not change. So

$$v = \text{rk}(N) = \text{rk}(N') = b - c + 1$$

□

Remark 8.8. This bound is tight for affine planes.

Theorem 8.9. Let $\mathcal{D}$ be a 2-design with a partition into c 1-designs. If $b = v + c - 1$, two blocks in the same class meet in $r - \lambda - k$ points; two blocks in different classes meet in k^2/v points.

9 Affine Planes

We generalize our construction of affine planes (from points and lines in $\mathbb{F}^2$). Work with $\mathbb{F}^{2d}$. Let $S_1, \dots, S_{q^d+1}$ be d -dimensional subspaces of $\mathbb{F}^{2d}$ such that $S_i \cap S_j = \{0\}, \forall i, j$. Our design has the elements of $\mathbb{F}^{2d}$ as points and the cosets of $S_1, \dots, S_{q^d+1}$ as its lines. It is a $(q^{2d}, q^d, 1)$ -design. Also, it is an affine plane. It is not necessarily isomorphic to our usual “desarguesian” plane. It is a translation plane.

The sets $S_1, \dots, S_{q^d+1}$ are pairwise skew subspaces. Note that $|S_i| = q^d$, $|\mathbb{F}^{2d}| = q^{2d}$, and

$$\frac{q^{2d} - 1}{q^d - 1} = q^d + 1$$

We call such a set of subspaces a spread. We can specify the subspaces by $2d \times d$ matrices, with column space $S_i, i = 1, \dots, q^d + 1$. Denote these matrices by M_i . When are the $2d \times d$ matrices

$$\begin{bmatrix} A \\ B \end{bmatrix}, \begin{bmatrix} C \\ D \end{bmatrix}$$

skew? If

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

is invertible.

Notice if Q is $d \times d$ invertible, then

$$\begin{bmatrix} A \\ B \end{bmatrix} \& \begin{bmatrix} A \\ B \end{bmatrix}^Q = \begin{bmatrix} AQ \\ BQ \end{bmatrix}$$

has the same column spaces. Next, $\begin{bmatrix} A \\ B \end{bmatrix}$ and $\begin{bmatrix} I \\ 0 \end{bmatrix}$ are skew if

$$\begin{bmatrix} I & A \\ 0 & B \end{bmatrix}$$

is invertible. That is, if B is invertible.

Also, $\begin{bmatrix} A \\ B \end{bmatrix}$ and $\begin{bmatrix} 0 \\ I \end{bmatrix}$ skew if

$$\begin{bmatrix} A & 0 \\ B & I \end{bmatrix}$$

is invertible, that is, if A is invertible. If A^{-1} exists, then

$$\text{col} \begin{bmatrix} A \\ B \end{bmatrix} = \text{col} \begin{bmatrix} I \\ BA^{-1} \end{bmatrix}$$

When are $\begin{bmatrix} I \\ A \end{bmatrix}$ and $\begin{bmatrix} I \\ B \end{bmatrix}$ skew?

$$\begin{bmatrix} I & I \\ A & B \end{bmatrix} \approx \begin{bmatrix} I & 0 \\ A & B - A \end{bmatrix}$$

Hence $B - A$ must be invertible. So our spread is given matrices

$$\begin{bmatrix} 0 \\ I \end{bmatrix}, \begin{bmatrix} I \\ 0 \end{bmatrix}, \begin{bmatrix} I \\ M_1 \end{bmatrix}, \dots, \begin{bmatrix} I \\ M_{q^d-1} \end{bmatrix}$$

where M_i is invertible and $M_i - M_j$ is invertible.

If we have a collection of $d \times d$ matrices $L_1, \dots, L_m$ such that $L_i - L_j$ is invertible, how large can m be? The answer is q^d .

9.1 An affine plane of order nine

Let $U = \mathbb{F}_3^2$. We want a spread in $V \oplus V$. This will be given by a set of 10 matrices:

$$V(0), V(\infty), M_1, \dots, M_8$$

where $M_i - M_j$ is invertible if $i \neq j$. The matrices are 2×2 over $\mathbb{F}_3$. Let $SL(2, 3)$ be the group of invertible 2×2 matrices of $\mathbb{F}_3$ with determinant one. What is the order of $SL(2, 3)$? Start with $GL(2, 3)$, there are 8 choices for the first row since it cannot be zero, and 6 choices for the second row since it is linearly independent to the first, so there are 48 choices. Since $\det : GL(2, 3) \rightarrow \{\pm 1\}$, so $|SL(2, 3)| = 24$. Let Σ be a Sylow 2-subgroup of $SL(2, 3)$ with order 8. We work out the elements of order two in Σ . If $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$\alpha^2 = \begin{pmatrix} a^2 + bc & (a+d)b \\ c(a+d) & d^2 + bc \end{pmatrix}$$

So if α has order two, then the off-diagonal entries must be zero. So $b = c = 0$ or $a + d = 0$. In the latter case $\alpha^2 = \begin{pmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{pmatrix} = I$ therefore $a^2 + bc = 1$. However, $\det(\alpha) = ad - bc = -a^2 - bc = 1$, contradicts. So we must have $b = c = 0$. Therefore $\alpha = -I$. That is, we have a unique element of order two in Σ .

We claim that $-I$ acts fixed point freely on $\mathbb{F}_3^2$. Consequently each element of Σ acts fixed-point freely. If $\alpha, \beta \in \Sigma$ and $\alpha^{-1}\beta \cdot u \neq u$, then $\beta u \neq \alpha u$ and $(\beta - \alpha)u \neq 0$ for any u . Since this holds for any $u \neq 0$, we have $\beta - \alpha$ is invertible. A plane constructed from a spread is called a translation plane.

10 Distance-regular graphs

If u is a vertex in the graph X , the distance partition with respect to X has i -cell equal to the vertices at distance i from, $\forall i$.

Definition 10.1 (according to Wiki). A graph X is distance regular if there are integers $p_{i,j}(r)$ such that each pair of vertices at distance r has exactly $p_{i,j}(r)$ common neighbours.

Definition 10.2 (AGT). A graph X is distance regular if there are integers $a_i, b_i, c_i (i = 0, \dots, d)$ such that for each vertex u , any vertex at distance i from u has c_i neighbours that at distance $i-1$ from u , a_i neighbours that at distance i from u , and b_i neighbours that at distance $i+1$ from u .

Definition 10.3. Let u be a vertex in X and let π be the distance partition with respect to u . Then X is distance-regular if, for u , this partition is equitable.

There are matrix versions of the first and second definitions. If $\text{diam}(X) = d$, we define graphs $X_0, X_1, \dots, X_d$ (distance graphs). Here $V(X_i) = V(X)$, and $x \sim y$ in X_i iff $\text{dist}_X(x, y) = i$. Let $A_i = A(X_i), A_0 = I$. Note that $A_0 + \dots + A_d = J$. Definition (1) gives

$$A_i A_j = \sum_r p_{i,j}(r) A_r, \forall i, j$$

Definition (2) gives

$$A_1 A_i = c_i A_{i-1} + a_i A_i + b_i A_{i+1}, \forall i$$

This implies that $A_i = p(A_1)$ for some polynomial of degree i .

Example 10.4. 1. $K_n, K_{n,n}$

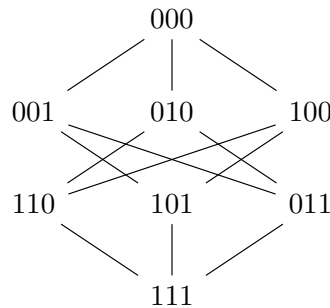
2. strongly regular graphs

3. cycles

Definition 10.5. A distance-regular graph is primitive if $X_1, \dots, X_d$ are connected.

Theorem 10.6 (Smith). If X is an imprimitive distance-regular graph, then either X_2 is not connected or X_d is not. In the first case, X is bipartite. In the second case, we say it is antipodal. (the relations “are-equal or at distance d ” is an equivalence relation)

Example 10.7.



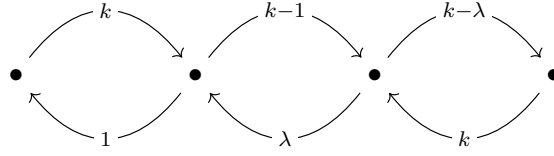
This is the 3-cube. We can partition the vertices set with distance 3.

The line graph of Petersen graph is distance-regular and antipodal.

11 Back to Symmetric Designs

Lemma 11.1. Suppose $\mathcal{D}$ is a symmetric $2-(v, k, \lambda)$ design. Then the incidence graph of $\mathcal{D}$ is distance-regular, bipartite, and has diameter 3.

Proof. distance diagram:



We can view it as a weighted directed graph.

$$\begin{pmatrix} 0 & k & 0 & 0 \\ 1 & 0 & k-1 & 0 \\ 0 & \lambda & 0 & k-\lambda \\ 0 & 0 & k & 0 \end{pmatrix}$$

The eigenvalues of this are the distinct eigenvalues of the incidence graph. □

If N is the incidence matrix of $\mathcal{D}$, then $A_0 = I$,

$$A_1 = \begin{bmatrix} 0 & N \\ N^T & 0 \end{bmatrix},$$

$$A_1^2 = \begin{bmatrix} NN^T & 0 \\ 0 & N^T N \end{bmatrix} = kI + \lambda \underbrace{\begin{bmatrix} J-I & 0 \\ 0 & J-I \end{bmatrix}}_{A_2}$$

Hence

$$A_3 = \begin{bmatrix} 0 & J-I-N \\ J-I-N^T & 0 \end{bmatrix}$$

This is also the incidence matrix of the complementary design.

The converse is true. If X is bipartite, distance regular, and of diameter three, it is the incidence graph of a symmetric 2-design.

12 Another section on the Affine planes

Consider the incidence graph of a projective plane of order n . The graph incidences by the points and vertices not on p or l is bipartite, distance-regular with diameter 4. It is antipodal.

12.1 affine plane minus parallel class

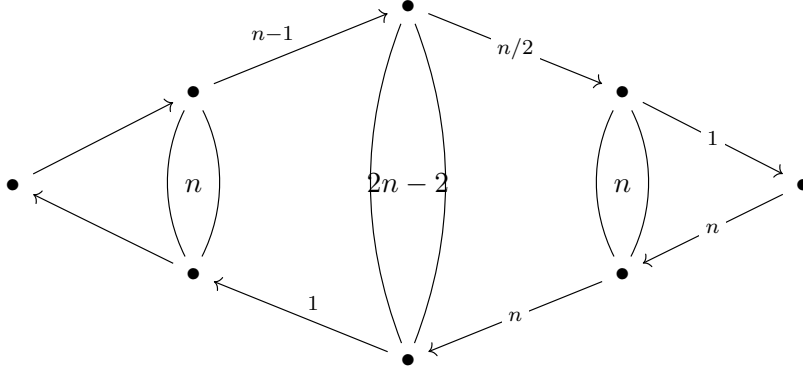
We are interested in the graph of affine plane minus parallel class, which is a graph on $2q^2$ vertices. It is distance regular, antipodal, with diameter 4. Indeed, it is K_{q^2, q^2} . We take (x, y) and $[a, b]$ in the two parallel classes, and we have $(x, y) \sim (a, b) \iff y = ax + b$

13 Hadamard graphs

Definition 13.1. $H = H_0 - H_1$, where H_0, H_1 are 01-matrices. $Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then we get the corresponding Hadamard graph

$$\begin{bmatrix} 0 & H_0 \otimes I + H_1 \otimes Q \\ H_0^T \otimes Q + H_1^T \otimes Q & 0 \end{bmatrix}$$

It is distance regular:



We can compute the eigenvalues of this graph: let $L = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then $LI_2L^T = I_2$ and $L \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} L^T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. So

$$\begin{bmatrix} I_n \otimes L & 0 \\ 0 & I_n \otimes L \end{bmatrix} \begin{bmatrix} 0 & H_0 \otimes I + H_1 \otimes Q \\ H_0^T \otimes Q + H_1^T \otimes Q & 0 \end{bmatrix} \begin{bmatrix} I_n \otimes L & 0 \\ 0 & I_n \otimes L \end{bmatrix}$$

gives

$$\begin{bmatrix} 0 & H_0 \otimes I_2 + H_1 \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ (H_0 \otimes I_2 + H_1 \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix})^T & 0 \end{bmatrix}$$

Which is permutation equivalent to

$$\begin{bmatrix} 0 & H_0 + H_1 & 0 & 0 \\ H_0^T + H_1^T & 0 & 0 & 0 \\ 0 & 0 & H_0 + H_1 & 0 \\ 0 & H_0^T + H_1^T & 0 & 0 \end{bmatrix}$$

Eigenvalues are $-n^{(4)}, -\sqrt{n}^{(n-1)}, 0^{(1)}$

14 t -designs

An incidence structure is a t -design if there are constants $\lambda_0, \dots, \lambda_k$ such that if $s \leq t$, then each set of s points lies in exactly λ_s blocks. For t -(v, k, λ) design, $\lambda = \lambda_t$. So λ_0 is the number of blocks, λ_1 is r . We say a t -design is a Steiner system if $\lambda = 1$. We already have a family of 3-designs. Let H be an $n \times n$ with first row equal to $\mathbf{1}^T$. Then each other row has the same number of 1's and -1's, therefore the row sum is zero. So we get $2n - 2$ blocks, in pairs, each block has size $\frac{n}{2}$. You may verify that this is a 3-($v, \frac{v}{2}, \frac{v}{4} - 1$) design.

n	parameters	
8	3-(8,4,1)	
12	3-(12,6,2)	Hadamard 2-design
16	3-(16,8,3)	
$\vdots$	$\vdots$	

If x is a point in a t design, then the set $\beta \setminus x$, where β is a block on x , form a $(t - 1)$ design on $V \setminus x$ with parameters

$$(t - 1)\text{-(}v - 1, k - 1, \lambda_{k-1}\text{)}$$

(which we call it the derived design). The block off x form a

$$(t - 1)\text{-(}v - 1, k, \lambda_{k-1}\text{)}$$

design.

Lemma 14.1. If $\mathcal{D}$ is a 3-(v, k, λ) design, $b \geq 2v - 2$. If equality holds, it is a Hadamard 3-design.

of the bound. Assume $x \in V$. Then blocks on x form a 2-design on $v - 1$ points, and so there are at least $v - 1$ blocks on x . Similarly the blocks off x give a 2-design on $v - 1$ points, and so there are at least $v - 1$ blocks on x . So $b \geq 2v - 2$. $\square$

15 Derived designs

If $\mathcal{D}$ is a t -design with $t \geq 2$, then its derived design is a $(t - 1)$ -design. If $\mathcal{D}'$ is the derived design of $\mathcal{D}$, we can attempt to construct $\mathcal{D}$ from $\mathcal{D}'$ by

- (a) adjoining a new point ∞ to each block
- (b) constructing blocks of size $k + 1$ from the points of V . [magic]

Note that $r_{\mathcal{D}} = b_{\mathcal{D}'}$, and equivalently, $(\lambda_1)_{\mathcal{D}} = (\lambda_0)_{\mathcal{D}'}$. In general, if $i > 0$, then $(\lambda_i)_{\mathcal{D}} = (\lambda_{i-1})_{\mathcal{D}'}$.

Lemma 15.1. The number of blocks in an extension of a t -(v, k, λ) design is

$$b \frac{v + 1}{k + 1}.$$

If $\mathcal{D}$ is a Steiner triple system, an extension $\mathcal{D}'$ of $\mathcal{D}$ would be a 3- $(v, 4, 1)$ design. The number of blocks would be

$$\frac{v(v-1)}{6} \cdot \frac{v+1}{4} = \frac{(v+1)v(v-1)}{24}$$

and this is an integer if $v \equiv 1, 3 \pmod{6}$.

It is a nontrivial result that every Steiner triple system admits an extension.

Example 15.2. 2-(7,3,1) gives 3-(8,4,1), which is a Hadamard 3-design.

2-(9,3,1) gives 3-(10,4,1)

Lemma 15.3. If a projective plane of order n has an extension, then $n \in \{2, 4, 10\}$.

Proof. The number of blocks in an extension would be

$$\frac{(n^2 + n + 1)(n^2 + n + 2)}{n + 2}$$

Then the remainder would be $((-2)^2 + (-2) + 1)((-2)^2 + (-2) + 2) = 12$. So

$$\frac{12}{n+2} \in \mathbb{N} \implies n \in \{2, 4, 10\}$$

□

When $n = 2$, we get a Hadamard design. When $n = 4$, 2-(21,5,1) gives 3-(22,6,1) design.

16 Möbius planes

If $\mathcal{D}$ is an affine plane with parameters $(n^2, n, 1)$, an extension would have

$$\frac{(n^2 + n)(n^2 + 1)}{n + 1} = n^3 + n$$

points. We want 3- $(n^2 + 1, n + 1, 1)$ designs.

A construction: Having a sphere on a plane, and say the north pole is p . Then p maps a circle inside the sphere to the plane (projecting points). This presents a 3-design as a 1-point extension of an affine plane.

How do we construct finite designs?

Let $\mathbb{F}$ be a finite field of order q and let $\mathbb{E}$ be a quadratic extension of $\mathbb{F}$. Let $\sigma : \mathbb{E} \rightarrow \mathbb{E}$ given by $x^\sigma := x^q$. If $\text{char}(\mathbb{F}) = q$, then

$$(x + y)^p = x^p + y^p$$

So $x \mapsto x^p$ is an automorphism. Thus σ is an action of $\mathbb{E}$ fixing each element in $\mathbb{F}$. Now

$$x^{\sigma^2} = (x^q)^q = x^{q^2} = x$$

and therefore σ is an involution. Now we work over $\mathbb{E}$. If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, a, b, c, d \in \mathbb{E}$$

then define $\tau_A : \mathbb{E} \cup \{\infty\} \rightarrow \mathbb{E} \cup \{\infty\}$ via

$$\tau_A(x) = \frac{ax + b}{cx + d}$$

a linear fractional map. We have

$$\tau_A(\infty) = \frac{a\infty + b}{c\infty + d} = \frac{a\infty}{c\infty} = \frac{a}{c}$$

Note that $\tau_{\gamma A} = \tau_A$, for all $\gamma \neq 0$. Moreover, we have $\tau_A \tau_B = \tau_{AB}$, and this is associative. This means we are working with $PGL(2, \mathbb{E})$.

Orders We have

$$|PGL(2, \mathbb{F})| = \frac{(q^2 - 1)(q^2 - q)}{q - 1} = q^3 - q$$

$$|PGL(2, \mathbb{E})| = q^6 - q^2$$

and

$$|PGL(2, \mathbb{E}) : PGL(2, \mathbb{F})| = q^3 + q$$

Theorem 16.1. Let $\mathbb{F}$ be a field of order q and $\mathbb{E}$ a quadratic extension of $\mathbb{F}$. Then the image of $\mathbb{F} \cup \infty$ under the action of $PGL(2, \mathbb{E})$ form a $3-(q^2 + 1, q + 1, 1)$ plane (A Möbius plane).

Theorem 16.2. Let $\mathcal{D}$ be a $3-(v, 4, 1)$ design with point set V and assume $V_1 \subseteq V$ with $v_1 = |V_1|, v = |V|$. If some subset of the blocks of $\mathcal{D}$ form a 3-design on V_1 , then $|V| \geq 2|V_1|$. If the equality holds, any block of $\mathcal{D}$ meets V_1 in an even number of points, and the blocks disjoint from V_1 form a 3-design.

Proof. Let $V_2 = V \setminus V_1$. Assume $y \in V_1$ and $x \in V_2$. If $z \in V_1 \setminus \{y\}$, then $\{y, z, x\}$ lies in a unique block $\{x, y, z, z'\}$, where $z' \in V_2$ (otherwise $\{y, z, z'\}$ lies in a block in V_1). Thus we have a bijection from $V_1 \setminus \{y\}$ to $V_2 \setminus \{x\}$, and so $v \geq 2v_1$.

Now, assume $v = 2v_1$. We show each block of $\mathcal{D}$ meets V_1 in an even number of points. If α is a block in $\mathcal{D}$ and $|\alpha \cap V_1| \geq 3$, the fourth point of α must lie in V_1 .

Now we show the blocks on V_2 form a 3-design. Assume $y \in V_1$, then y lies in $\frac{(v-1)(v-2)}{6}$ blocks of $\mathcal{D}$, and in $\frac{(v_1-1)(v_1-2)}{6}$ blocks in V_1 . Here

$$\frac{(v_1 - 1)(v_2 - 2)}{6} = \frac{(v - 2)(v - 4)}{24}$$

and so the number of blocks on y not in $\mathcal{D}_1$ is

$$\frac{(v - 1)(v - 2)}{6} - \frac{(v - 2)(v - 4)}{24} = \frac{v^2 - 2v}{8}$$

As before, for each $z \in V_1 \setminus \{y\}$ and each $x \in V_2$, there is a point z' in V_2 such that $\{x, y, z, z'\}$ is a block of $\mathcal{D}$. So there are exactly

$$\frac{1}{2} \left(\frac{v}{2} - 1 \right) \frac{v}{2} = \frac{v^2 - 2v}{8}$$

blocks on y meeting V_1 in exactly two points.

Finally, if u, v, w are distinct points in V_2 , they lie in a block of $\mathcal{D}$ disjoint from V_1 , which gives a 3-design on V_1 . $\square$

17 Incidence matrices (of subsets)

Let V be a set of size v . Define the matrix $W_{t,k}(v)$ be the 01-matrix with:

- rows indexed by t -subsets of V ,
- columns indexed by k -subsets of V ,
- $(W_{t,k}(v))_{\alpha,\beta} = 1$ if $|\alpha| = t, |\beta| = k, \alpha \subseteq \beta$.

Hence $W_{t,k}(v)$ is $\binom{v}{t} \times \binom{v}{k}$ with row sums equal to $\binom{v-t}{k-t}$ and the column sums $\binom{k}{t}$. When v is obvious, we may write $W_{t,k}$. We have

$$W_{t,t} = I_t$$

and $W_{1,t}(v)$ is the incidence matrix of the complete design with parameters $2-(v, k, \binom{v-2}{k-2})$.

Lemma 17.1. Let $\mathcal{B}$ be a subset of $\binom{V}{k}$ (set of all k -subsets of V), with characteristic vector z . Then $\mathcal{B}$ is a t -design if and only if

$$W_{t,k}z = \lambda_t \mathbf{1}$$

for some constant λ_t .

Proof. Exercise. $\square$

Lemma 17.2. $W_{s,t}W_{t,k} = \binom{k-s}{t-s}W_{s,k}$

Proof. Looking at $(W_{s,t}W_{t,k})_{\alpha,\beta}$ with $\alpha \subset \beta, |\alpha| = s, |\beta| = k$. Then we are just choosing t -subset containing α and contained in β . So there are $\binom{k-s}{t-s}$ of such. $\square$

Corollary 17.3. row space of $(W_{s,k}) \subseteq$ row space of $W_{t,k}$.

Lemma 17.4. If $\mathcal{D}$ is a t -design and $s \leq t$, then $\mathcal{D}$ is an s -design.

Proof. Assume $W_{t,k}z = \lambda_t \mathbf{1}$, then

$$\binom{k-s}{t-s}W_{s,k}z = W_{s,t}W_{t,k}z = \lambda_t W_{s,t}\mathbf{1} = \lambda_t \binom{v-s}{k-s}$$

and so if $s \leq t$, $\mathcal{D}$ is an s -design. $\square$

17.1 Complements

New incidence matrices

$$(\overline{W}_{t,k})_{\alpha,\beta} = \begin{cases} 1 & , \alpha \cap \beta = \emptyset \\ 0 & , \text{o/w} \end{cases}$$

$\overline{W}_{t,k}$ can be obtained by a permutation of the columns of $W_{t,N-k}$.

Lemma 17.5. (a) $\overline{W}_{i,k} \overline{W}_{t,k}^T = \binom{v-t-i}{k-t} \overline{W}_{i,t}$,

$$(b) \quad W_{i,k} \overline{W}_{i,k}^T = \binom{v-k-i}{k-i} \overline{W}_{i,k}$$

Proof. Exercise. □

Lemma 17.6. (a)

$$\overline{W}_{t,k} = \sum_i (-1)^i W_{i,t}^T W_{i,k}$$

(b)

$$W_{t,k} = \sum_i (-1)^i W_{i,t}^T \overline{W}_{i,k}$$

Proof. For (a), suppose α is a t -subset of V and β is a k -subset. Then $(\overline{W}_{t,k})_{\alpha,\beta} = 1$ if $\alpha \cap \beta = \emptyset$, and otherwise 0. We have

$$(W_{i,t}^T W_{i,k})_{\alpha,\beta} = \binom{|\alpha \cap \beta|}{i}$$

and the $\alpha\beta$ -entry of the sum in (b) is

$$\sum_i (-1)^i \binom{|\alpha \cap \beta|}{i} = \begin{cases} 1 & , \alpha \cap \beta = \emptyset; \\ 0 & , \text{o/w}. \end{cases}$$

□

Lemma 17.7. If $t \leq k \leq v - t$, then $\overline{W}_{t,k}$ and $W_{t,k}$ have the same rank over $\mathbb{Q}$.

Proof. We have $W_{t,k} = \binom{k-i}{t-i}^{-1} W_{i,t} W_{t,k}$ and so (a) in the previous lemma implies that

$$\overline{W}_{t,k} = \sum_i (-1)^i \binom{k-i}{t-i}^{-1} W_{i,t}^T W_{i,t} W_{t,k}$$

So each row of $\overline{W}_{t,k}$ is a linear combination of rows of $W_{t,k}$. Therefore $\text{row}(\overline{W}_{t,k}) \subseteq \text{row}(W_{t,k})$. We also have

$$W_{i,t} \overline{W}_{t,k} = \binom{v-k-i}{t-i} W_{i,k}$$

and as $W_{t,k} = \sum_i (-1)^i W_{i,t}^T \overline{W}_{i,k}$ □

Theorem 17.8. $\text{rk}(W_{t,k}) = \min\{\binom{v}{t}, \binom{v}{k}\}$.

Proof. Assume $t \leq k \leq v - t$. First, assume that $v - t = k$. Then $W_{t,k}$ and $\overline{W}_{t,k}$ are squares of the same order. But $\overline{W}_{t,v-t}$ is a permutation matrix and is therefore invertible. Since $\overline{W}_{t,v-t}$ and $W_{t,v-t}$ have the same row space, $W_{t,v-t}$ is invertible.

Now if $t \leq h \leq v - t$, then

$$W_{t,h}W_{h,v-t} = \binom{v-2t}{h-t} W_{t,v-t}$$

Since the RHS is invertible, the rows of $W_{t,h}$ are linearly independent. $\square$

Lemma 17.9. The complement of a t -design is a t -design.

Proof. From above, there are matrices G and H , with constant row sums, such that

$$\overline{W}_{t,k} = GW_{t,k}, W_{t,k} = H\overline{W}_{t,k}$$

Hence if $W_{t,k}x = \lambda \mathbf{1}$, then

$$\overline{W}_{t,k}x = \lambda G\mathbf{1} = c\mathbf{1}.$$

Since $W_{t,v-k}$ is gotten from $\overline{W}_{t,k}$ by permuting columns, it follows that the complement of a t -design is a t -design. $\square$

Definition 17.10. Let M be a $m \times n$ matrix. The characteristic matrix of a partition $\rho = \{\rho_1, \dots, \rho_h\}$ is the matrix with the characteristic vectors of the cells of ρ as its columns. If ρ is a partition of the rows of M with char matrix R . This gives a partition of the columns of M (by equality of columns of $R^T M$). We call this the induced partition and denote it by ρ^* .

Theorem 17.11. Assume M is an $m \times n$ matrix with linearly independent rows. If ρ is a partition of the rows of M , then $|\rho| \leq |\rho^*|$.

Proof. Let R be the char matrix of ρ . Then $|\rho^*|$ is equal to the number of distinct columns of $R^T M$, and so

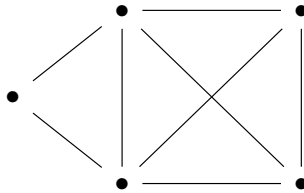
$$\text{rk}(R^T M) \leq |\rho^*|$$

If $x^T R^T M = 0$, then $x^T R^T = 0$ (rows of M are linearly independent). So $x^T = 0$ (rows of R^T are linearly independent). So the rows of $R^T M$ are linearly independent, and hence $\text{rk}(R^T M) = |\rho|$. Therefore $|\rho| \leq |\rho^*|$. $\square$

17.2 Application in Graphs

Definition 17.12 (graph). A graph is a set V of vertices and a set E of edges, where each edge is an unordered pair of vertices.

Is the following a graph?



NO! This is an “unlabelled graph”.

The number of graphs on n vertices is $2^{\binom{n}{2}}$.

Claim: Let $Gr(n, e)$ denote the number of isomorphism classes of graphs on n vertices with edges. Then if $e \leq \binom{n}{2}$, then

$$Gr(n, e-1) \leq Gr(n, e)$$

Proof. Consider $W(e-1, e)$, then the rows gives ρ and columns have at least ρ^* cells. □

Lemma 17.13. If $\mathcal{D}$ is a symmetric design and π is an automorphism of $\mathcal{D}$, the number of points fixed by π equals the number of blocks.

Proof. Let N be the incidence matrix of $\mathcal{D}$. If π is represented by the pair of permutations (P, Q) , then $PNQ^T = N$. So

$$PN = NQ$$

and, as N is invertible,

$$N^{-1}PN = Q$$

and

$$\# \text{ points fixed by } P = \text{tr}(P) = \text{tr}(N^{-1}PN) = \text{tr}(Q)$$

□

Theorem 17.14. If $s \leq t$ and $(V, \mathcal{B})$ is a t -(v, k, λ) design, then

$$\binom{v}{s} \lambda_s = b \binom{k}{s} \quad \binom{v}{t} \lambda_t = b \binom{k}{t}$$

and therefore

$$\frac{\binom{v}{t} \lambda_t}{\binom{v}{s} \lambda_s} = \frac{\binom{k}{t}}{\binom{k}{s}}$$

implying that

$$\frac{\lambda_t}{\lambda_s} = \frac{\binom{v-t}{k-t}}{\binom{v-s}{k-s}}.$$

Theorem 17.15. If $(V, \mathcal{B})$ is a t -design and $i + j \leq t$, then

$$|\mathcal{B}|^{-1} W_i(\mathcal{B}) W_j(\mathcal{B})^T = \frac{1}{\binom{v}{k}} W_{i,k} W_{j,k}^T$$

Proof. Notice $|\mathcal{B}| = \lambda_0$, so we have

$$(|\mathcal{B}|^{-1} W_i(\mathcal{B}) W_j(\mathcal{B})^T)_{\alpha, \beta} = \frac{\lambda_{i+j}}{\lambda_0} = \binom{v-i-j}{k-i-j} \binom{v}{k}^{-1} = \frac{1}{\binom{v}{k}} W_{i,k} W_{j,k}^T$$

□

Theorem 17.16 (Ray-Chaudhuri & Wilson). If $\mathcal{D}$ is a t -design with blocks, then

$$b \geq \binom{v}{\lfloor \frac{t}{2} \rfloor}.$$

Proof. Let M be the incidence matrix for $\lfloor \frac{t}{2} \rfloor$ -subsets of V versus k -subsubsets. Set $s = \lfloor \frac{t}{2} \rfloor$. Then

$$\frac{1}{|\mathcal{B}|} W_s(\mathcal{B}) W_s(\mathcal{B})^T = \frac{1}{\binom{v}{k}} W_{s,k} W_{s,k}^T$$

Notice that the RHS is invertible, so it follows that the rows of $W_s(B)$ are linearly independent. $\square$

Remark 17.17. If $b = \binom{v}{\lfloor \frac{t}{2} \rfloor}$, we say $\mathcal{D}$ is a tight design (so symmetric designs are tight).

18 Polynomial Spaces

Let Ω be the set of all k -subsets of $V = \{1, \dots, v\}$. For all $i \in V$, define functions

$$f_i(\beta) = \begin{cases} 1 & , i \in \beta \\ 0 & , \text{o/w} \end{cases}$$

Let $Pol(\Omega, 0)$ be constant functions, $Pol(\Omega, 1) = Pol(\Omega, 0) \cup \{f_i : i \in V\}$ linear functions, inductively define $Pol(\Omega, r+1) = Pol(\Omega, r) \circ Pol(\Omega, 1)$.

18.1 Averages

If $f \in Pol(\Omega)$, define

$$\langle 1, f \rangle := \frac{1}{|\Omega|} \sum_{\alpha \in \Omega} f(\alpha)$$

and, more generally,

$$\langle f, g \rangle = \langle 1, fg \rangle = \frac{1}{|\Omega|} \sum_{\alpha \in \Omega} f(\alpha)g(\alpha)$$

A subset Φ of Ω is a t -design if whenever $f \in Pol(\Omega, t)$, we have

$$\frac{1}{|\Phi|} \sum_{\alpha \in \Phi} f(\alpha) = \langle 1, f \rangle$$

Or in another word,

$$\langle 1, f \rangle_\Phi = \langle 1, f \rangle_\Omega$$

Theorem 18.1. A subset Φ of Ω is a t -design if and only if it is a t -design in our usual sense.

Theorem 18.2 (Ray-Chaudhuri & Wilson). If Φ is a t -design, then $|\Phi| \geq \dim Pol(\Omega, \lfloor \frac{t}{2} \rfloor)$.

Proof. Assume $\dim(\text{Pol}(\Omega, \lfloor \frac{t}{2} \rfloor)) = m$ and let $f_1, \dots, f_m$ be an orthonormal basis for $\text{Pol}(\Omega, \lfloor \frac{t}{2} \rfloor)$. For all i, j , we have $f_i f_j \in \text{Pol}(\Omega, t)$, and so

$$\delta_{ij} = \langle f_i, f_j \rangle = \langle 1, f_i f_j \rangle = \langle 1, f_i f_j \rangle_\Phi = \langle f_i, f_j \rangle_\Phi$$

So the restriction $f_i \in \Phi$ are orthogonal and hence linearly independent on Φ , therefore

$$m \leq |\Phi|$$

□

19 Codes and Designs

Definition 19.1. A code is a subset of $V(X)$, for some graph X . (The elements of the code are code words). The set of vertices at distance r from vertex u is $B_r(u)$, the ball of radius r about u . The packing radius of a code $\mathcal{C}$ is the maximum value of e such that the balls of radius about words in $\mathcal{C}$ are pairwise disjoint. The covering radius is the minimum r such that the balls of radius r about words in $\mathcal{C}$ cover $V(X)$.

All our codes will be subsets of $H(n, q)$ (think of $H(n, q)$ as the n -dim vector space over $GF(q)$).

Definition 19.2. A code is perfect if its packing and covering radius are equal. For a perfect e -code, the balls of radius e are pairwise disjoint.

Lemma 19.3. If $H(n, q) = X$, then we have

$$|B_r(n)| = 1 + n(q-1) + \dots + \binom{n}{r}(q-1)^r.$$

Lemma 19.4. If the packing radius is e , then

$$|\mathcal{C}| \cdot |B_e(a)| \leq q^n;$$

if the covering radius is r , then

$$|\mathcal{C}| \cdot |B_r(u)| \geq q^n$$

The first inequality implies “sphere-packing bound”:

$$|\mathcal{C}| \leq \frac{q^n}{|B_e(u)|}$$

Example 19.5. Assume X is a k -regular graph, So if $u \in V(X)$, then $|B_1(u)| = 1 + k$. Hence if $\mathcal{C}$ is a code with packing radius 1, then

$$|\mathcal{C}|(k+1) \leq |V(X)|$$

What if $\mathcal{C}$ is perfect? If x is a char vector of $\mathcal{C}$, then

$$Ax = ax + b(\mathbf{1} - x), A\mathbf{1} = k\mathbf{1}$$

So $\text{span}\{x, \mathbf{1}\}$ is A -invariant.

Lemma 19.6. If the regular graph X has a perfect 1-code, then -1 is an eigenvalue of X .

19.1 Linear Codes

Definition 19.7. A linear code is a subspace of $V(n, q)$. So the row space of a matrix over $\mathbb{F}_q$ is a linear code. A generator matrix for a linear code $\mathcal{C}$ is a $d \times n$ matrix G such that $\text{row}(G) = \mathcal{C}$ and the rows of G are linearly independent. If G is the generator matrix for $\mathcal{C}$ on the map

$$a^T \mapsto a^T G$$

is the “encoding” of a . We say G is in standard form if

$$G = [I_d \ G_0]$$

If $\mathcal{C}$ is a code, then

$$\mathcal{C}^\perp = \{z : \langle z, c \rangle = 0, c \in \mathcal{C}\}$$

is the dual code to $\mathcal{C}$. A matrix H is a check matrix for $\mathcal{C}$ if $(\ker(H))^T = \mathcal{C}$. If $G = [I_d \ G_0]$ is a generator matrix, then $H = [-G_0^T \ I_{n-d}]$ is a check matrix.

$$GH^T = [I_d \ G_0] \begin{bmatrix} -G_0 \\ I_{n-d} \end{bmatrix} = 0$$

A vector z^T is a code word if $H z = 0$.

Definition 19.8. A code is self-orthogonal if $\mathcal{C} \subseteq \mathcal{C}^\perp$ and is self-dual if $\mathcal{C} = \mathcal{C}^\perp$

Lemma 19.9. If G is a generator matrix for $\mathcal{C}$ over $GF(2)$, and $\mathcal{C}$ is self-orthogonal, and all rows of G have even weight (Hamming distance to the 0 vector). Then all words in $\mathcal{C}$ have even weight. If rows of G are doubly even (all words have weights divisible by 4), and $\mathcal{C}$ is self-dual, then all words in $\mathcal{C}$ are doubly even

Proof. Exercise. □

Lemma 19.10. If $\mathcal{C}$ is a linear code of length n and dimension d , then $\mathcal{C}^T$ has length n and

$$\dim(\mathcal{C}^T) = n - \dim(\mathcal{C})$$

The check matrix is a generator matrix for the dual code. If $\mathcal{C}$ is self-dual, then $\dim(\mathcal{C}) = \frac{n}{2}$.

Lemma 19.11. If $U \leq V$ (U is a subspace of V), then

$$\dim(U) + \dim(U^\perp) = \dim(V)$$

and

$$(U^\perp)^\perp = U$$

It does NOT follow that $U \cap U^\perp = \langle 0 \rangle$. We can have $U \leq U^\perp$.

19.2 Codes from Designs

Definition 19.12. The order of a $2-(v, k, \lambda)$ design is $r - \lambda$.

Lemma 19.13. Let $\mathcal{D}$ be a 2-design with incidence matrix N and order n . If p is a prime and $p \nmid n$, then $\text{rk}_p(N) \geq v - 1$. If $p \mid n$, then $\text{rk}_p(N) \leq \frac{v+1}{2}$.

Proof. We have

$$NN^T = nI + \lambda J$$

If $p \nmid n$, then $\text{rk}_p(nI) = v$ and $\text{rk}_p(N) \geq p - 1$. If $p \mid n$, then $\text{rk}_p(NN^T) \leq 1 \implies \text{rk}(N) \leq \frac{v+1}{2}$. $\square$

Claim: If H is a 12×12 Hadamard matrix, then $\text{rk}_3(H) = 6$.

We work over $\mathbb{Z}$. If $a, b \in \mathbb{Z}$, and $\gcd(a, b) = d$, then exists x, y such that $xa + yb = d$, and so

$$\underbrace{\begin{pmatrix} x & y \\ -\frac{b}{d} & \frac{a}{d} \end{pmatrix}}_{\det=1} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} d \\ 0 \end{pmatrix}$$

A matrix over $\mathbb{Z}$ is unimodular if its determinant is ± 1 , and unimodular matrices form a group with matrix multiplication.

Lemma 19.14. If N is $m \times n$ over $\mathbb{Z}$, there exists unimodular matrices L and R such that $LN R$ is diagonal.

Lemma 19.15. Let M be an $m \times n$ matrix over $\mathbb{Z}$. If $p^l \parallel \det(M)$, then $\text{rk}_p(M) \geq n - l$.

Proof. Choose unimodular L, R such that $LMR = D$, D diagonal. Since L, R are unimodular, so $\text{rk}_p(M) = \text{rk}_p(D)$. Hence at most l diagonal entries of D are divisible by p , and so $\text{rk}_p(M) \geq n - l$. $\square$

Example 19.16. Let H be a 12×12 Hadamard matrix. Then $HH^T = 12I$ and so $\det(H)^2 = 12^{12}$. Therefore $\det(H) = 12^6 = 2^{12} \cdot 3^6$, and $\text{rk}_3(H) \leq 12 - \text{rk}_3(H) \implies \text{rk}_3(H) \leq 6$. By the lemma, we have equality.

Note: we have a $2-(7, 3, 1)$. It has extensions to a $3-(8, 4, 1)$ design (Hadamard). We also have a $2-(9, 3, 1)$ design (affine plane), with extensions $3-(10, 4, 1)$ (Möbius), and then extend to $4-(11, 5, 1)$, and then extend to $5-(12, 6, 1)$.

19.3 Codes & Designs

If N is an incidence matrix of a design $\mathcal{D}$, and p is a prime, there are three codes over $GF(p)$ associated with N :

- (a) $C_p(\mathcal{D})$: spanned by the rows of N^T . (the rows of N^T are just columns of N , which is the char vector of the blocks)
- (b) Hull: $H_p(\mathcal{D}) = C_p(\mathcal{D}) \cap C_p(\mathcal{D})^\perp$.
- (c) $E_p(\mathcal{D})$: spanned by the differences of the rows of N^T . (The codimension of $E_p(\mathcal{D})$ in $C_p(\mathcal{D})$ is at most one)

If N is the incidence matrix of a symmetric $2-(v, k, \lambda)$ design, then

$$NN^T = nI + J$$

If p is a prime and $p|n$, then

$$(e_i - e_j)^T NN^T \equiv (e_i - e_j)^T J = 0 \pmod{p}$$

And so $(e_i - e_j)^T NN^T (e_k - e_l) = 0$.

19.4 Minimum Weight

A block β is tangent to a subset S of $V(\mathcal{D})$ if $|\beta \cap S| = 1$. A vector x lies in $\mathcal{C}^\perp$ if $\langle x, \beta \rangle = 0$ for all blocks β . So if S is tangent to a block, its char function does not lie in $\mathcal{C}^\perp$.

Two special cases:

1. $\min \text{wt} = 1$ (in $\mathcal{C}^\perp$). Then the first column of $\mathcal{C}$ is zero.
2. $\min \text{wt} = 2$, redundant.

Lemma 19.17. If $\mathcal{C} = C_p(\mathcal{D})$, for a $2-(v, k, \lambda)$ design, the minimum weight of $\mathcal{C}$ is at least $\frac{r+\lambda}{\lambda}$.

Proof. Let V be the point set of $\mathcal{D}$ and suppose $S \subseteq V$. If $i \in S$, at most $\lambda(|S| - 1)$ blocks contain i and another point of S . So if $r > (|S| - 1)\lambda$, there is a block β that is tangent to S . Therefore, no subset of size less than $(r + \lambda)/\lambda$ can be the support of a word. $\square$

Lemma 19.18. Let $\mathcal{D}$ be a symmetric $2-(v, k, \lambda)$ design with the point set V . Let p be a prime such that $p|n$. If $S \subset V$ and $|S| = k$, and the char vector of S lies in $\mathcal{C}$, then S is a block.

Proof. Let $z(U)$ denote the char vector of a subset U of V . We prove first that if $z(S) \in \mathcal{C}$, then for any blocks β , we have $|S \cap \beta| \geq \lambda$. Assume u and w are vectors in $\mathcal{C}$ and

$$u = Nx, w = Ny \text{ for some } x, y$$

As $p|n$, we have $NN^T = kJ \pmod{p}$ and

$$w^T n = y^T N^T N x = ky^T J x = k^2 \mathbf{1}^T y \mathbf{1}^T x \pmod{p}$$

and since $k\mathbf{1} = N\mathbf{1}$,

$$w^T u = k \mathbf{1}^T x \mathbf{1}^T y = \mathbf{1}^T u \mathbf{1}^T w$$

Hence if u and v are char vectors of k -subsets,

$$kw^T u = k^2$$

and so $w^T u = k \pmod{p}$. So $w^T u = \lambda \pmod{p}$. Next we use this to show that S is a block. Set $z = z(S)$, and $y = N^T z$. If $\hat{y} = y - \lambda \mathbf{1}$, then $\hat{y} \geq 0$ (exercise) and it follows that

$$\langle \mathbf{1}, \hat{y} \rangle = n, \langle \hat{y}, \hat{y} \rangle = n^2$$

which implies one entry of $\hat{y}$ is n and other entries are zero. Therefore, S is a block. $\square$

20 Ovals in projective planes

Definition 20.1. An arc in a projective plane is a set of points, no three in a line. An s-arc has exactly s points.

Example 20.2. In the plane from $V(3, \mathbb{F}_q)$. Some points $\begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix}, t \in \mathbb{F}_q$. Any three of these points are linearly independent.

If $a = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$, then $a^T \begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix} = a_0 + a_1 t + a_2 t^2$. These points, together with $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ (think about $1, \infty, \infty^2$), this gives $q + 1$ points, which form an arc.

Lemma 20.3. The maximum number of points on an arc in a projective plane of order q is $q + 2$. If the equality holds, q is even.

Proof. Let S be an arc and p be a point on S . Then picking another point p' on the arc gives a unique line through p and p' by the definition of arc. So

$$|S| - 1 \leq q + 1$$

We now consider the equality: let y be a point not on S . Each line y lies in must meet S in exactly two points (if exactly 1, then y can be included in S , if more than 2, then S is not an arc). Hence the lines on y partition S into two parts, therefore $|S|$ is even and hence q is even. $\square$

Definition 20.4. A $(q + 1)$ -arc is called an oval, a $(q + 2)$ -arc is a hyperoval.

Lemma 20.5. If q is even, the tangents to a $(q + 1)$ -arc are concurrent.

Proof. Let P, Q be distinct points on an oval. There is a unique tangent to the oval at each point on it. Hence each point of $P \vee Q$ (the line joins P and Q) lies on a unique tangent. Consequently, each point on a secant to the oval lies on a unique tangent. Assume K is the point of intersection of two distinct tangents. Then K cannot lie on a secant. Therefore all lines on K are tangents. $\square$

20.1 Ovals in the plane of order four

The number of blocks in an extension of a $2-(21, 5, 1)$ design is

$$b \frac{v + 1}{k + 1} = \frac{21 \times 22}{6} = 77$$

To construct the extension, we get 21 blocks by adding ∞ to each line. We need 56 5-sets to form the extra blocks in the extension 3-design.

20.2 Back to codes again

We work in a plane of order n with incidence matrix N . The code of the plane is the row space of N^T over $GF(p)$. It is spanned by the char vectors of the blocks. We assume that $p|n$.

Theorem 20.6. Let $\mathcal{P}$ be a projective plane of order n with incidence matrix N . If $n \equiv 2 \pmod{4}$, the extension of the code spanned by the rows of N^T is self-dual and doubly even.

20.3 Codes from projective planes

We work in planes of order n with the incidence matrix N . The code of the plane is the row space of N^T ; it is spanned by the char. vectors of the blocks. Assume $p|n$.

Theorem 20.7. Let $\mathcal{P}$ be a projective plane of order n with incidence matrix N . If $n \equiv 2 \pmod{4}$. The extension of the code of $\mathcal{P}$ spanned by the rows of N^T is self-dual and doubly even (each code has weight divisible by 4).

Proof. Set $\hat{N} = [N\mathbf{1}]$. So $\text{row}(\hat{N})$ is the extended code. We have

$$\hat{N}\hat{N}^T = NN^T + J = nI + 2J \equiv 0 \pmod{2}$$

and so the extended code is self-orthogonal. Since the weight of each row of $\hat{N}$ is divisible by four, the code is doubly even (exercise).

To prove the extended code is self-dual, we compute its dimension. Since $\hat{N}\hat{N}^T = 0 \pmod{2}$, we have

$$\text{rk}_2(\hat{N}) \leq \frac{1}{2}(n^2 + n + 2)$$

and, as $\mathbf{1} \in \text{col}(N)$, we have $\text{rk}_2(N) = \text{rk}_2(\hat{N})$. For the lower bound,

$$\det(NN^T) = (r - \lambda)^{v-1} \text{rk}(N)$$

(this holds for all 2-designs) and hence, for a projection plane,

$$\det(N) = n^{(N-1)/2}(n+1)$$

and therefore $2^{(N-1)/2}$ is the largest power of 2 that divides $\det(N)$. Accordingly, $\text{rk}_2(N) \geq \frac{v+1}{2}$. $\square$

20.4 Weight enumerators of codes

Let $\mathcal{C}$ be a code of length n and let A_i denote the number of code words of weight i in $\mathcal{C}$. The weight enumerator of $\mathcal{C}$ is

$$W_{\mathcal{C}}(x, y) := \sum_{i=0}^n A_i x^i y^{n-i}$$

We define two operations on codes:

1. The words of the punctured codes are those we get by deleting the i -th coordinate, we denote it by $\mathcal{C} \setminus i$.
2. The shortened code $\mathcal{C}/i$ is got by taking words in $\mathcal{C}$ with 0 in position i and then deleting the i -th coordinate.

So both $\mathcal{C} \setminus i$ and $\mathcal{C}/i$ have length $n - 1$.

Lemma 20.8. If $\mathcal{C}$ is a linear code, then

$$(\mathcal{C} \setminus i)^\perp = \mathcal{C}^\perp / i, (\mathcal{C}/i)^\perp = \mathcal{C}^\perp \setminus i$$

Proof. Exercise. □

Define i in $\{1, \dots, n\}$ to be a loop if $x_i = 0$ for all x in $\mathcal{C}$; i is a coloop if there is a word on $\mathcal{C}$ with support $\{i\}$. We see that i is a loop in $\mathcal{C}$ if and only if it is a coloop in $\mathcal{C}^\perp$.

Lemma 20.9. Let $\mathcal{C}$ be a linear code. If i is a loop in $\mathcal{C}$,

$$W_{\mathcal{C}}(x, y) = yW_{\mathcal{C} \setminus i}(x, y),$$

if it is a coloop,

$$W_{\mathcal{C}}(x, y) = (y + (q - 1)i)W_{\mathcal{C}/i}(x, y)$$

If i is neither a loop nor a coloop,

$$W_{\mathcal{C}}(x, y) = (y - x)W_{\mathcal{C}/i}(x, y) + xW_{\mathcal{C} \setminus i}(x, y)$$

Proof. The first two claims are exercises. Assume i is neither a loop nor a coloop, let D_0 be the code words zero on i and D_1 be the rest. Because i is not a loop, $D_1 \neq \emptyset$. The weight enumerator for D_0 is $yW_{\mathcal{C}/i}(x, y)$. Let φ be the map that sends a word x in $\mathcal{C}$ to the word we get by deleting the i -th coordinate. As i is not a coloop, $\ker(\varphi) = \langle 0 \rangle$ and φ is injective and maps words of wt r in D_1 to words of wt $r - 1$ in $(\mathcal{C} \setminus i) - (\mathcal{C}/i)$ and so

$$W_{D_1}(x, y) = x(W_{\mathcal{C} \setminus i}(x, y) - W_{\mathcal{C}/i}(x, y))$$

As $W_{\mathcal{C}} = W_{D_0} + W_{D_1}$, the lemma follows. □

Theorem 20.10 (MacWilliams). If $\mathcal{C}$ is a linear code over $GF(q)$, then

$$W_{\mathcal{C}^\perp}(x, y) = \frac{1}{|\mathcal{C}|} W_{\mathcal{C}}(y - x, y + (q - 1)x)$$

Proof. We prove it by induction on the length. The theorem holds for i that is a loop or coloop (exercise). If i is neither a loop nor a coloop, then

$$W_{\mathcal{C}^\perp}(x, y) = (y - x)W_{\mathcal{C}^\perp/i}(x, y) + xW_{\mathcal{C}^\perp \setminus i}(x, y)$$

The RHS here equals

$$(y - x)W_{(\mathcal{C}/i)^\perp}(x, y) + xW_{(\mathcal{C} \setminus i)^\perp}(x, y)$$

Assume $\dim(\mathcal{C}) = k$. Because i is not a coloop,

$$|\mathcal{C} \setminus i| = |\mathcal{C}| = q^k$$

By induction,

$$W_{\mathcal{C}^\perp}(x, y) = q^{-k}[(y - x)W_{\mathcal{C} \setminus i}(y - x, y + (q - 1)x) + qxW_{\mathcal{C}/i}(y - x, y + (q - 1)x)]$$

Since $qx = y + (q - 1)x - (y - x)$, the RHS is the image of

$$\frac{1}{|\mathcal{C}|} [(y - x)W_{\mathcal{C}/i}(x, y) + xW_{\mathcal{C} \setminus i}(x, y)]$$

under the map that sends $x \mapsto y$ and $y \mapsto y + (q - 1)x$. □

Theorem 20.11. If $n \equiv 6 \pmod{8}$, there is no projection plane of order n .

Proof. Assume $n \equiv 2 \pmod{4}$. Then the extended code of the plane is self-dual and doubly even. Since $\mathcal{C}$ is doubly even,

$$W_{\mathcal{C}}(ix, y) = W_{\mathcal{C}}(x, y)$$

As $\mathcal{C}$ is self-dual,

$$W_{\mathcal{C}}(x, y) = W_{\mathcal{C}^\perp}(x, y) = \frac{1}{|\mathcal{C}|} W_{\mathcal{C}}(-x + y, x + y) = W_{\mathcal{C}}\left(\frac{-x + y}{\sqrt{2}}, \frac{x + y}{\sqrt{2}}\right)$$

Thus, $W_{\mathcal{C}}(x, y)$ is invariant under the substitution:

$$x \mapsto ix, y \mapsto y \quad \sim \quad \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sigma_i : x \mapsto \frac{-x + y}{\sqrt{2}}, y \mapsto \frac{x + y}{\sqrt{2}} \quad \sim \quad \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

The composition of these is

$$\frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix}$$

Now,

$$\left(\frac{1}{\sqrt{2}} \begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix}\right)^3 = \frac{1+i}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

If $\tau = \frac{1+i}{\sqrt{2}}$, then $|\tau| = 8$ (order). Then

$$W_{\mathcal{C}}(x, y) = W_{\mathcal{C}}(\tau x, \tau y) = \tau^n W_{\mathcal{C}}(x, y) \implies 8|n$$

□

Definition 20.12 (Golay codes, Witt design, Mathien groups). Let H be a 12×12 Hadamard matrix. Then the row space of H over $GF(3)$ gives a code of length, with words of 0,6,9,12. The supports of the words of weight six are the blocks of a 5-(12,6,1) design.

$$5-(12,6,1) \rightarrow 4-(11,5,1) \rightarrow 3-(10,4,1) \text{ (Möbius plane)} \rightarrow 2-(9,3,1) \text{ (affine plane)}$$

Now, we construct a 5-(24,8,1) design. Let N be the incidence matrix of the symmetric 2-(11,6,3) design (the complement of the 2-(11,5,2) Hadamard design). Define

$$\hat{N} = \begin{bmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & N \end{bmatrix}, G = [I\hat{N}]$$

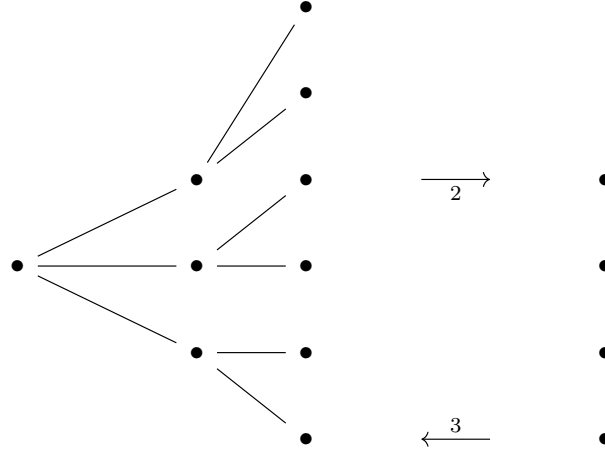
Lemma 20.13. The row space of $\mathcal{C}$ over $GF(12)$ is self-dual with minimum weight 8.

Theorem 20.14. The words of weight 8 are the blocks of a 5-(24,8,1) design.

$$5-(24,8,1) \rightarrow 4-(23,7,1) \rightarrow 3-(22,6,1) \rightarrow 2-(21,5,1)$$

20.5 Uniqueness of planes of order 2 and 4

(a) 2-(7,3,1): consider the incidence graph (so-called Heawood graph).



(b) 2-(21,5,1). An s -arc in a plane is a set of s points, no three in a line. A $(q+1)$ -arc in a plane of order q is an oval. If q is even, each oval lies in a $(q+2)$ -arc – a hyperoval.

Claim: A plane of order four has a hyperoval.

21 Difference sets

Definition 21.1. A linear space is an incidence structure where any two points lie on exactly one line.

Theorem 21.2 (De Brnijn & Erdős). Let $\mathcal{D}$ be a linear space on v points, then either $\mathcal{D}$ has exactly one line, or $\mathcal{D}$ has at least v lines. If equality holds, $\mathcal{D}$ is a dual linear space.

Proof. If x is a point of $\mathcal{D}$, let $r(x)$ denotes the number of lines on x , then $k(\beta)$ denotes the number of points on the block β . Assume there is more than one line. If x is not a point on β , then $r(x) \geq k(\beta)$. Let b be the number of lines in $\mathcal{D}$. Suppose that $b \leq v$. Then $vr(x) \geq bk(\beta)$ and hence

$$b(v - k(\beta)) \geq v(b - r(x)).$$

This implies that

$$1 = \sum_x \sum_{\beta} \frac{1}{v(b - r(x))} \geq \sum_{\beta} \sum_x \frac{1}{b(v - k(\beta))} \geq 1$$

Therefore we shall obtain the equality $b = v$. □

Definition 21.3. Let Γ be an abelian group, and let C be a subset of Γ . If $g \in \Gamma$, we define

$$C + g := \{c + g : c \in C\}.$$

This is the translate of C under g . We have incidence structures with Γ as the point set and the set of translates of C as the block set. The canonical example is a 2-(7,3,1) design:

$$C = \{0, 1, 3\}, \Gamma = \mathbb{Z}_7$$

Theorem 21.4. Let C with $|C| \geq 3$ be a subset of the abelian group Γ and suppose that for each non-zero $g \in \Gamma$, there is a unique pair (x, y) of elements of C such that $g = y - x$. Then the incidence structure of $(\Gamma, \{C + g : g \in \Gamma\})$ is a projective plane.

Proof. Suppose x, y are distinct elements of C and for some $h \in \Gamma$,

$$u = x + h, v = y + h$$

Then $v - u = y - x$ and, if the state condition on C holds, then any two translates of C have at most one point in common. We also see that if $v - u = y - x$, the translate $C + h$ contains u in v . It follows that each pair of elements of C lies in a translate of C and so our incidence structure is a linear space.

Since the number of translates of C is at most $|\Gamma|$, De Bruijn & Erdős implies the structure is a dual linear space. If $|C| \geq 3$, it is a projective plane. $\square$

21.1 Ovals and Difference Set

Definition 21.5. If D is a subset of an abelian group Γ , define

$$-D := \{-g : g \in \Gamma\}$$

Lemma 21.6. If D is a planar difference set (it gives a projective plane) in an abelian group, then $-D$ is an oval.

Proof. Let $\mathcal{P}$ be the plane generated by D . If S is a set of points in $\mathcal{P}$, then S is an arc if and only if for each translate of D of $\mathcal{P}$,

$$|S \cap (D + g)| \leq 2$$

If $-a, -b, -c \in -D$, and

$$-a = u + g, -b = v + g, -c = w + g, \text{ for some } u, v, w \in D$$

Then

$$-g = a + u = b + v$$

and so

$$a - b = v - u$$

As D is a planar difference set, it follows that

$$a = v \text{ \& } b = u$$

and hence

$$-g = b + v = c + w$$

Similarly, we have $b = v$ and $c = v$. This implies D is an oval. $\square$

21.2 Cayley Graphs

Let P be the permutation matrix with first row (01000000) and each row shifts right one digit. Set $M = I + P + P^3$. Then

$$(I + P + P^3)(I + P^{-1} + P^{-3}) = 3I + (P + P^2 + P^3 + P^{-1} + P^{-2} + P^{-3}) = 3I + (J - I)$$

A Cayley graph is obtained from a group Γ by choosing a subset of C of Γ and defining $X(\Gamma, C)$ to be the graph with vertex set Γ and with arc from g to h iff $hg^{-1} \in C$.

If $\Gamma = \mathbb{Z}_7$ and $C = \{0, 1, 3\}$, then $A(X) = I + P + P^3$.

21.3 Mutually unbiased bases

Definition 21.7. A complex Hadamard matrix is a flat unitary matrix. A complex Hadamard matrix has Butson index k if its entries are k -th roots of unity. (so real Hadamard matrices have Butson index 2)

Definition 21.8. A pair of orthonormal bases $x_1, \dots, x_d$ and $y_1, \dots, y_d$ (in $(\mathbb{C}^d)$) are unbiased if there is a constant $\gamma \in \mathbb{R}_{\geq 0}$ such that

$$\langle x_i, y_j \rangle \langle y_j, x_i \rangle = \gamma, \forall i, j$$

A set of orthonormal bases is mutually unbiased if each pair of distinct bases is unbiased.

The columns of a $d \times d$ unitary matrix are an orthonormal basis. The columns of unitary matrices U & V are unbiased if U^*V is flat.

Notice that U^*V is unitary and so U^*V is a flat unitary matrix.

Example 21.9. 1. $\{I, H\}$, where H is real-Hadamard.

$$2. \left\{ I_2, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \right\}$$

3. If U and V are flat, so is $U \otimes V$. If U and V are unitary, so is $U \otimes V$.

Exercise: If $M \otimes N$ is flat, then M and N are flat (Mullin).

If $(x_i)_i$ and $(y_j)_j$ are unbiased bases, then

$$y_j = \sum_i c_i x_i$$

for some c_i 's. Taking inner product of y_j and x_i gives c_i , that is $\langle y_j, x_i \rangle = c_i, \forall i, j$. Also,

$$1 = \langle y_j, y_j \rangle = \sum_{i,k} c_i c_k \langle x_i, x_k \rangle = \sum_i c_i^2 = d\gamma$$

Therefore we have $\gamma = \frac{1}{d}$.

21.4 Maximum size

How large can a set of mutually unbiased matrices be? If P and Q are projections onto lines in a set of mutually unbiased bases, then $P = u_i u_i^*$, $Q = v_k v_k^*$, and

$$\langle P, Q \rangle = \text{tr}(P^* Q) = \text{tr}(u_i u_i^* v_k v_k^*) = \begin{cases} 1 \\ 0 \\ \frac{1}{d} \end{cases}$$

21.5 Gram matrices

If $x_1, \dots, x_m$ are vectors in an inner product space, their Gram matrix is

$$G := (\langle x_i, x_j \rangle) = [x_1, \dots, x_m]^* [x_1, \dots, x_m]$$

So G is positive semidefinite:

$$u^* G u = u^* [x_1, \dots, x_m]^* [x_1, \dots, x_m] u = ([x_1, \dots, x_m] u)^* ([x_1, \dots, x_m] u) \geq 0$$

Also,

$$\dim(\text{span}\{x_1, \dots, x_m\}) = \text{rk}(G)$$

We have

$$G = I_m \otimes I_d + \frac{1}{d}(J_m - I_m) \otimes J_d$$

The eigenvalues of $(J_m - I) \otimes J_d$ are

e.val	multi
$(m-1)d$	1
0	$m(d-1)$
$-d$	$m-1$

Remark 21.10. 1. If $Hx = \lambda x$ and $Ky = \mu y$, then

$$(H \otimes K)(x \otimes y) = (Hx) \otimes (Ky) = (\lambda x) \otimes (\mu y) = \lambda \mu (x \otimes y)$$

2. e.vals of J_d are d (multi 1) and 0 (multi $d-1$).

Therefore the eigenvalues of $I_{md} + \frac{1}{d}(J_m - I_m) \otimes J_m$ are

e.vals	multi
m	1
1	$m(d-1)$
0	$(m-1)$

Therefore

$$\text{rk}(G) = 1 + m(d-1)$$

The space of $d \times d$ has dimension d^2 (over $\mathbb{R}$). So

$$\begin{aligned} 1 + m(d-1) &\leq d^2 \\ \implies m(d-1) &\leq d^2 - 1 \implies m \leq d+1 \end{aligned}$$

21.6 Equiangular lines

Let $x_1, \dots, x_m$ be unit vectors. $|\langle x_i, x_j \rangle|^2$ independent of i and j :

$$m \leq \begin{cases} \binom{d+1}{2} & \text{real} \\ d^2 & \text{complex} \end{cases}$$

21.7 Relative difference sets

Definition 21.11. Let G be an abelian group. We want to work in the group ring (algebra) of elements $\sum_{g \in G} c_g g, c_g \in \mathbb{F}$. If $S \subseteq G$, we may write S for $\sum_{g \in S} g$. A subset D of G is a difference set if

$$DD^{-1} = |D|1_G + \lambda(G \setminus \{1_G\})$$

That is, we can view the elements of DD^{-1} as $|G| \times |G|$ permutation matrices. Then a difference set gives rise to a symmetric design.

Suppose $N \triangleleft G$. Then a subset D of G is a relative difference set is

$$DD^{-1} = |D|1_G + \lambda(G \setminus N)$$

Remark 21.12. $DD^{-1} = \sum_{g,h \in D} gh^{-1}$. There could be repetitions. If $|N| = n, |D| = k, |G| = mn$, then the definition gives a (m, n, k, λ) difference set.

Definition 21.13. A relative difference set is regular if $m = k$.

Theorem 21.14. The existence of a real difference set with parameters (k, n, k, λ) implies the existence of a set of $n + 1$ MUBs in $\mathbb{C}^k$. (So (k, k, k, λ) gives a complete set).

21.8 Affine planes

Let $V = GF(q^2)$, with elements (x, y) . Let $[a, b]$ denote the set

$$\{(x, y) : y = ax + b\}$$

We get a bipartite graph with the elements (x, y) of V as points, $[a, b]$ as lines and edges:

$$(x, y) \sim [a, b] \iff y = ax + b$$

This graph is distance regular with diameter four; being equal or at distance four is an equivalence relation on the vertices. The automorphism group of this graph contains an abelian group acting regularly on each colour class. Using this, we can determine the eigenvectors. Then the restrictions of the eigenvectors to the neighbourhood of a vertex form a complete set of mutually unbiased bases. We don't actually need finite fields for this – we can use commutative semifields. It follows that if q is a prime power, there is a complete set of $q + 1$ MUBs in $\mathbb{C}^q$. There is a product construction. If $H_1, \dots, H_m$ and $K_1, \dots, K_m$ give MUBs, then the matrices

$$H_i \otimes K_i \quad (i = 1, \dots, m)$$

form a set of MUBs.

If

$$n = p_1^{\mu_1} \cdots p_k^{\mu_k}$$

and $\mu = \min\{\mu_i\}$, then there is a set of $\mu + 1$ MUBs in $\mathbb{C}^n$.

This is somewhat construction. We can get six MUBs in $\mathbb{C}^{26^2}$ (The product only gives five). The construction uses orthogonal arrays. In particular an $OA(26, 6)$.